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(54) **RELOAD SHAFT ASSEMBLY FOR SURGICAL STAPLER**

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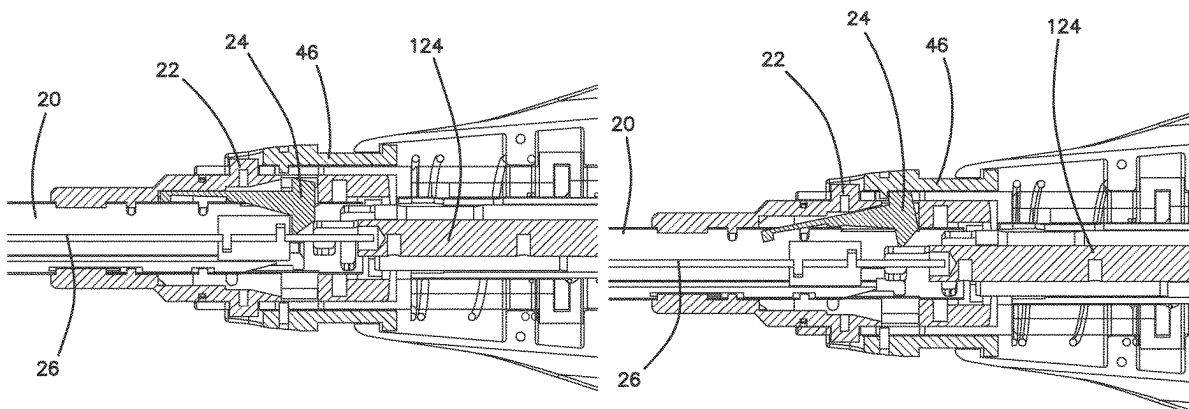
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(57) **ABSTRACT**
A surgical stapling system can include a reload shaft. The shaft can include an elongate tubular member with have a jaw assembly at the distal end thereof and a coupling collar at the proximal end thereof. The shaft assembly also includes an articulation joint coupling the jaw assembly to the distal end. A drive member and an articulation member extend within the tubular body of the shaft from the proximal end to the distal end. A firing member is connected to the distal end of the drive member such that advancement of the drive beam advances the firing member to close the jaw
(Continued)



assemblies and fire staples from a reload positioned in the jaw assembly. The shaft assembly can also include a lockout mechanism to prevent a firing operation on a previously-fired reload or no reload.

9 Claims, 26 Drawing Sheets

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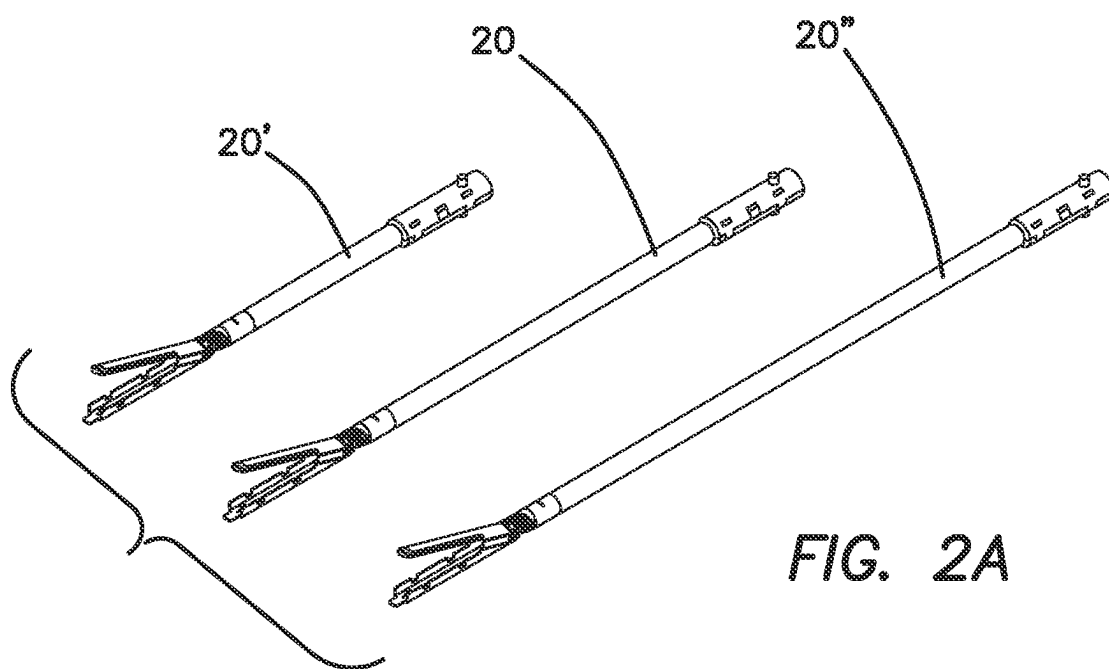
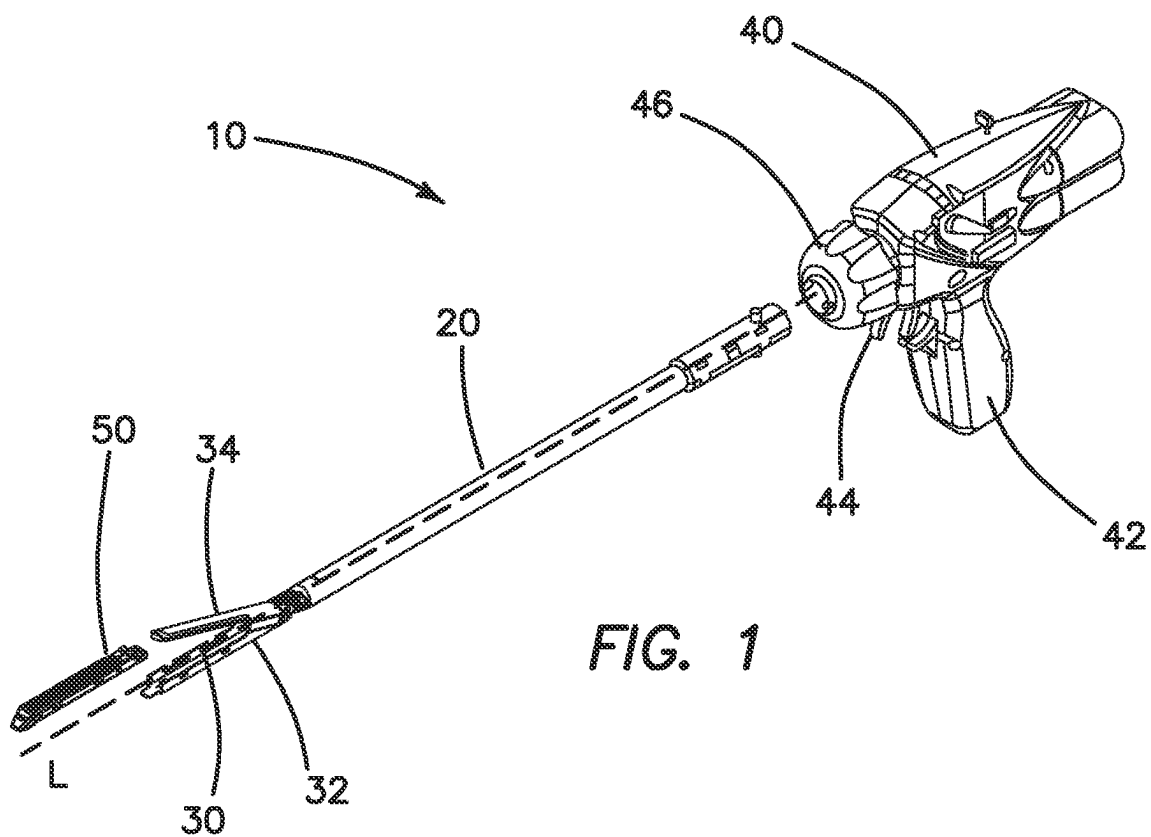
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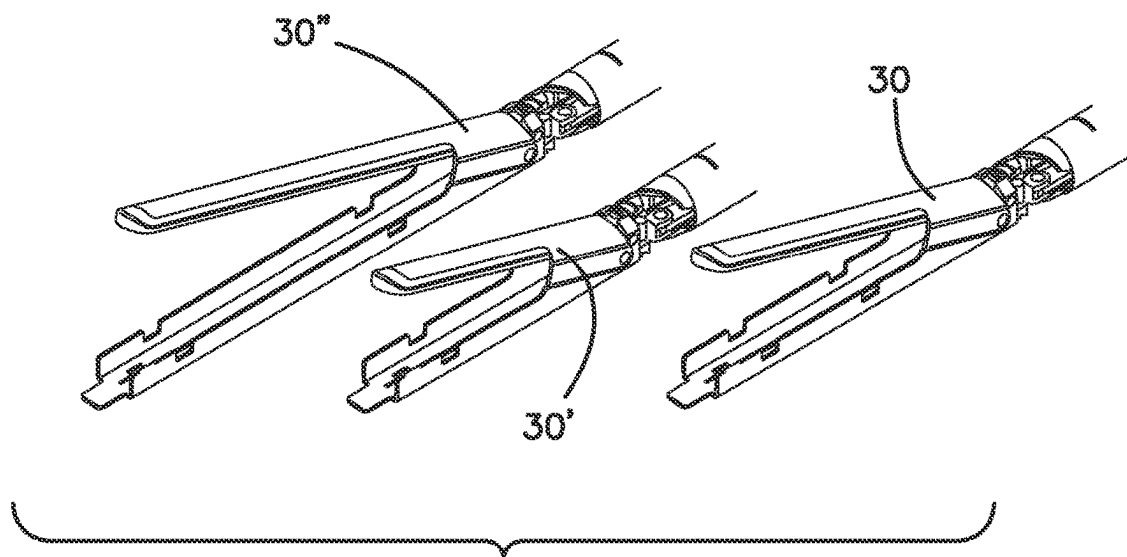


FIG. 2B

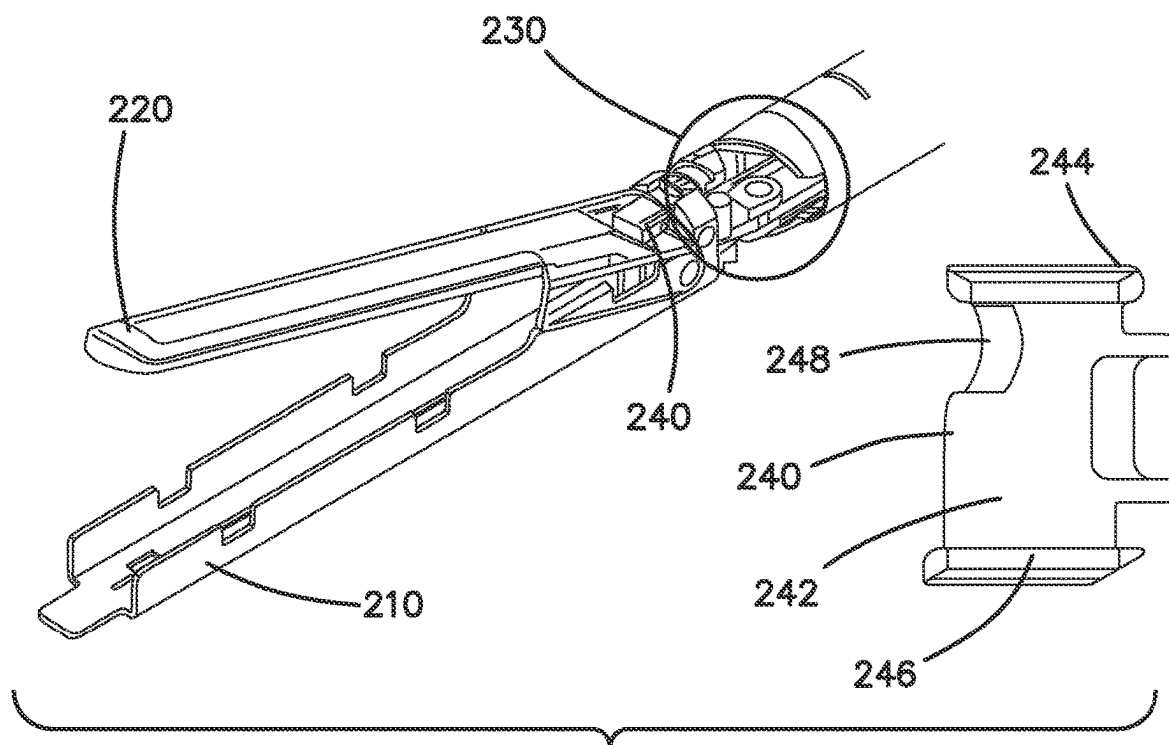
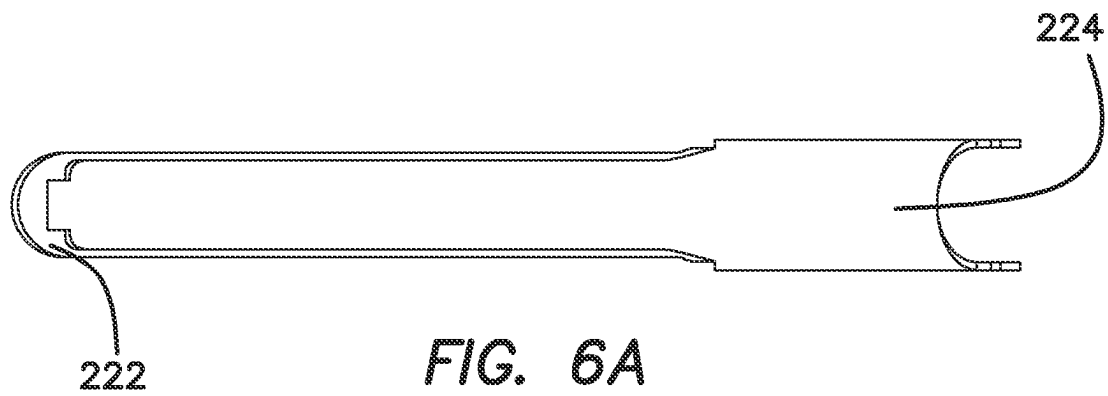
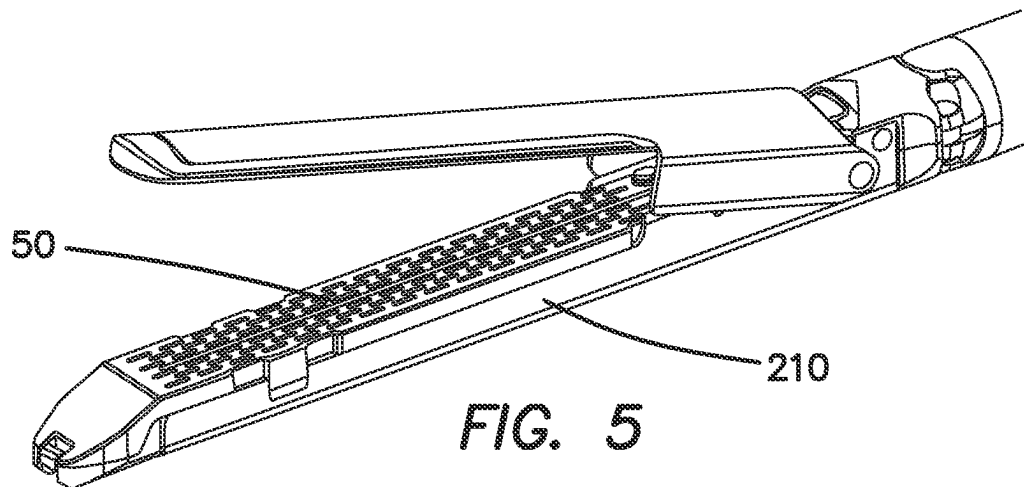
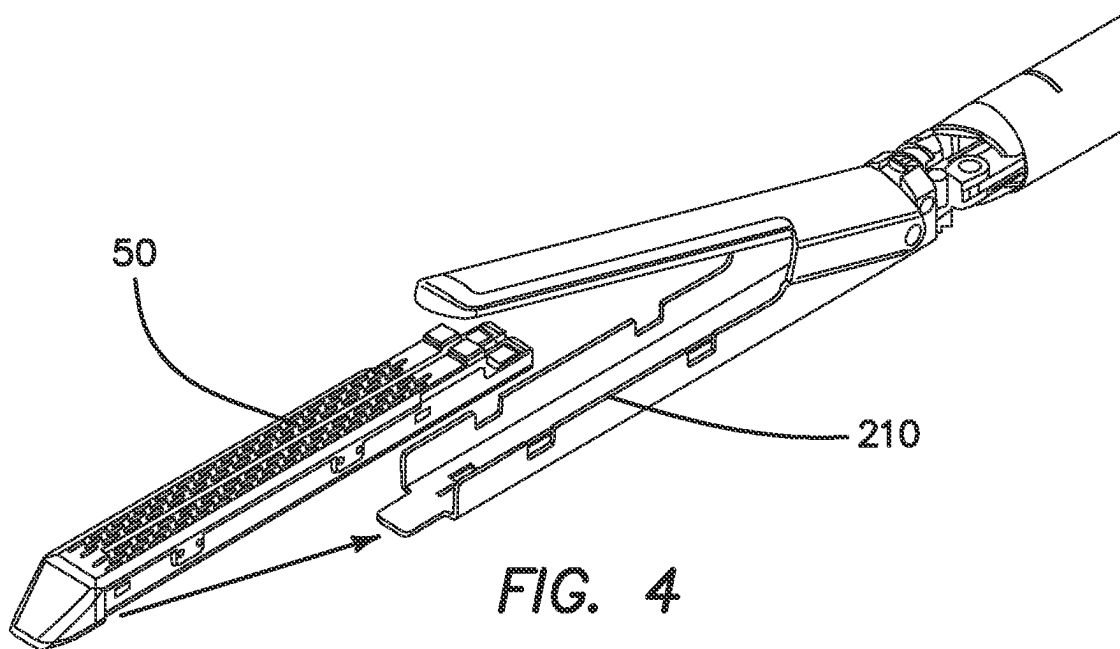


FIG. 3



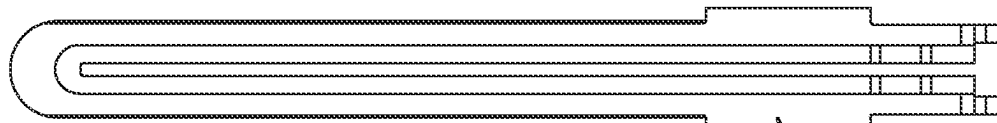


FIG. 6B

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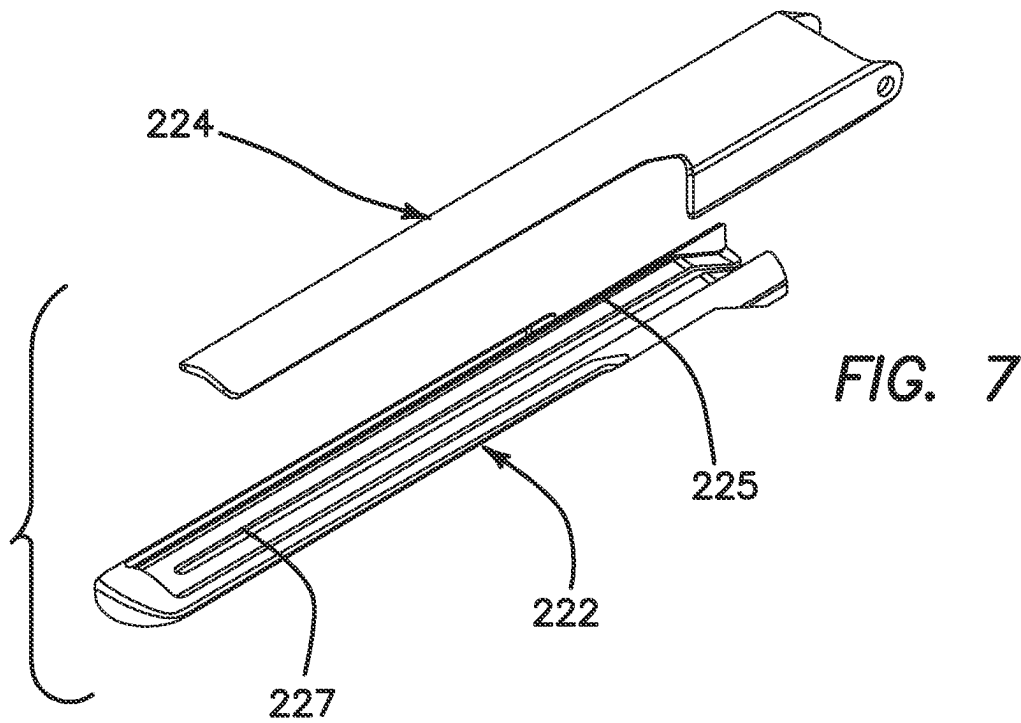


FIG. 7

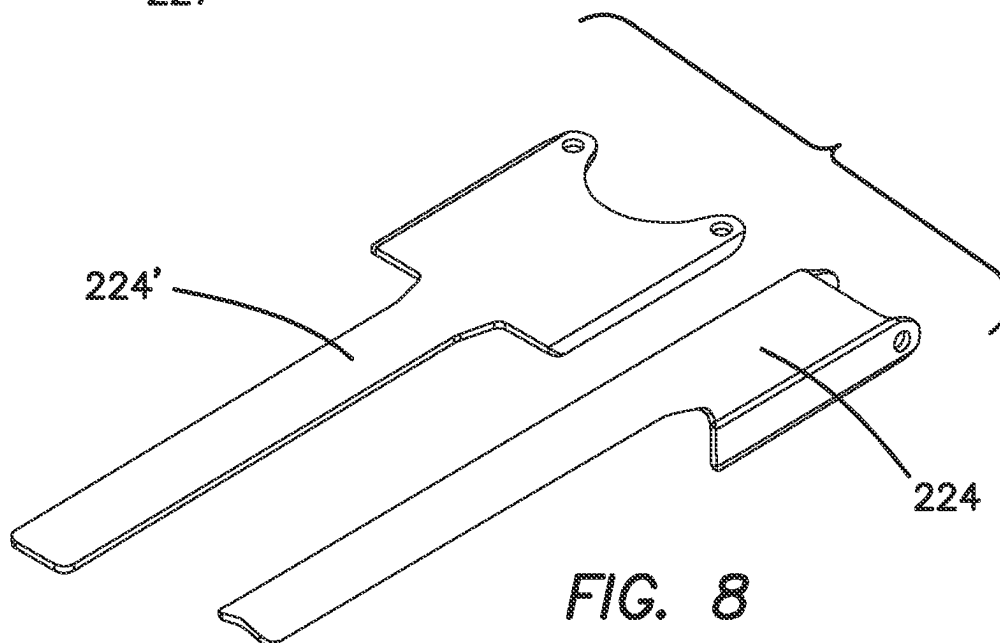


FIG. 8

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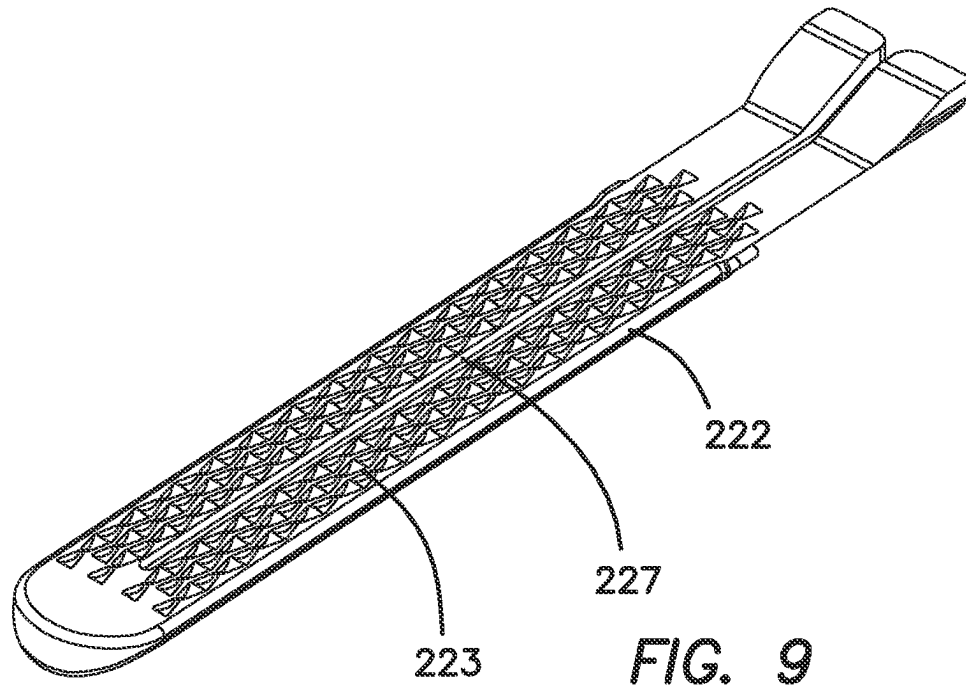


FIG. 9

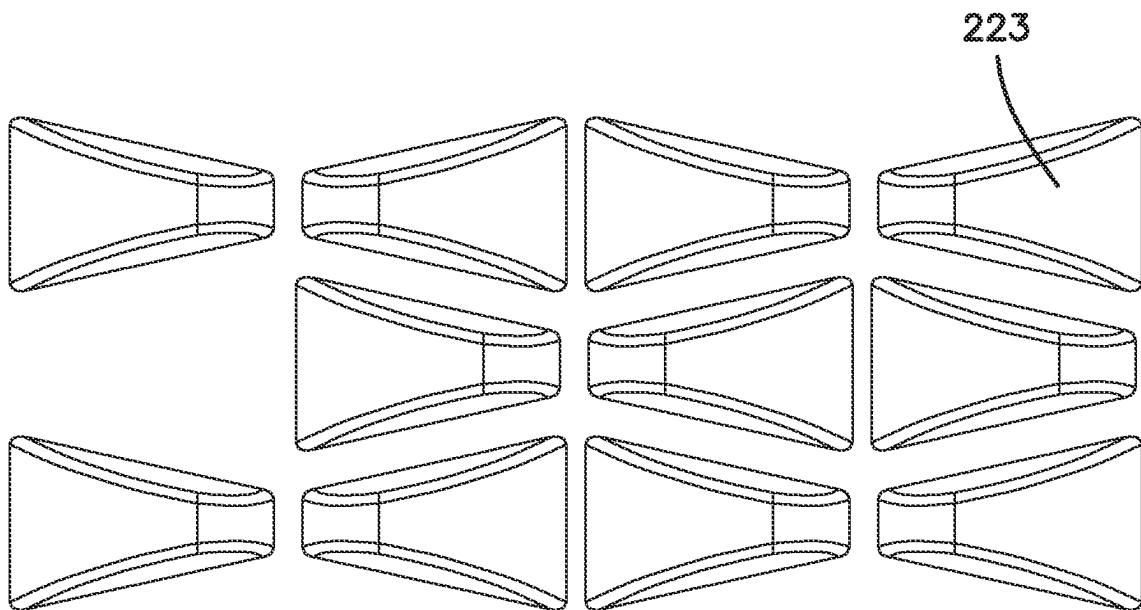
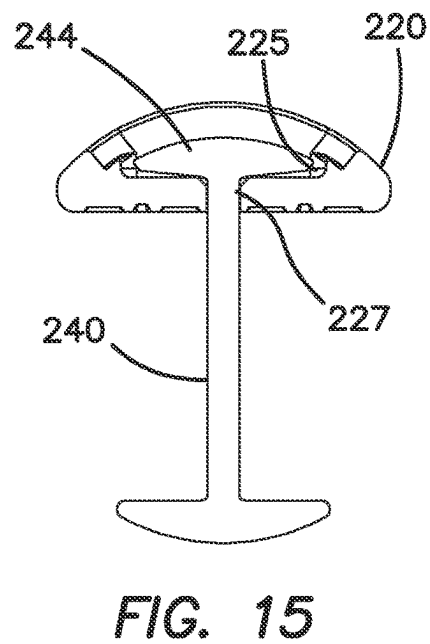
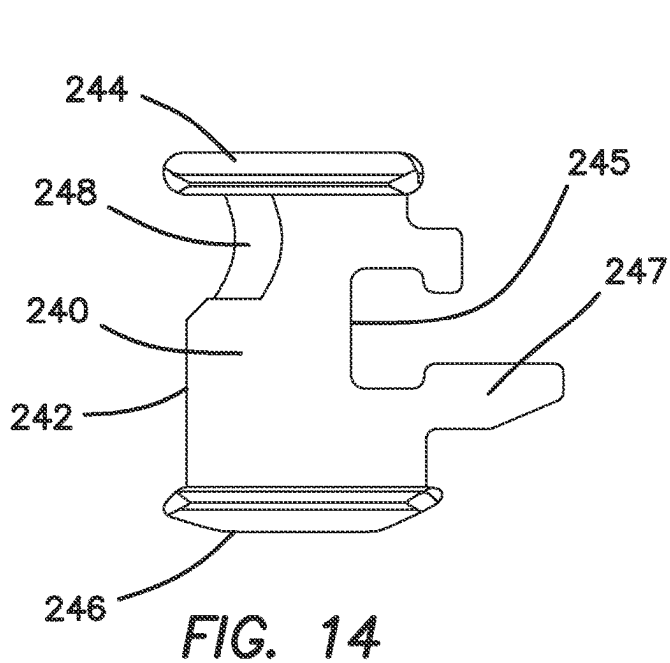
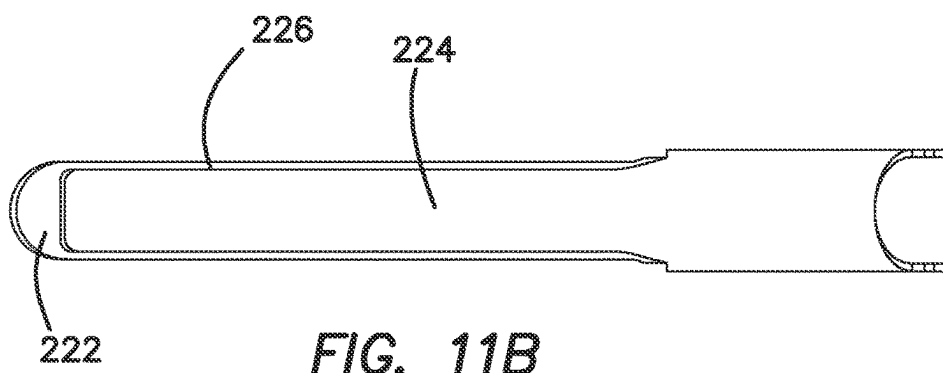
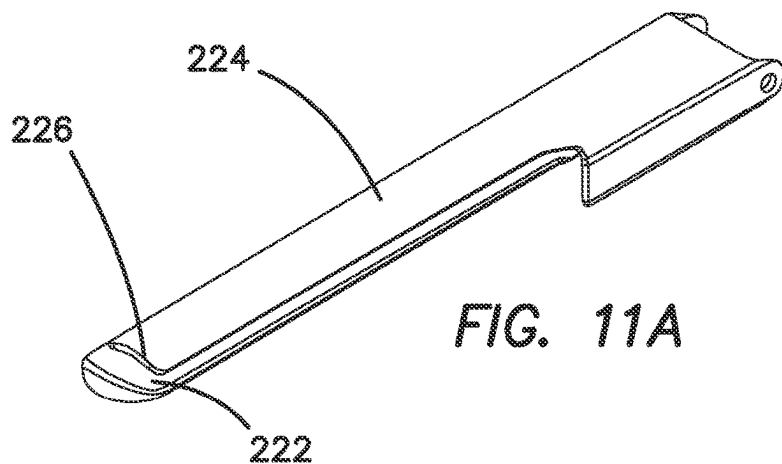


FIG. 10



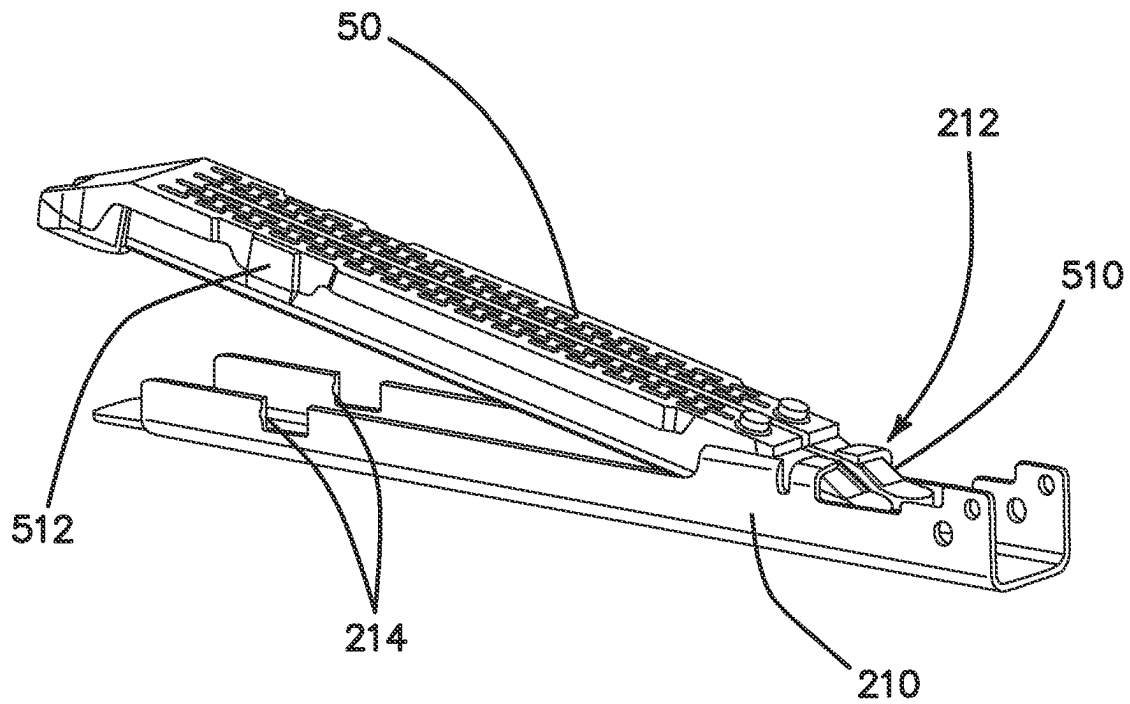


FIG. 12

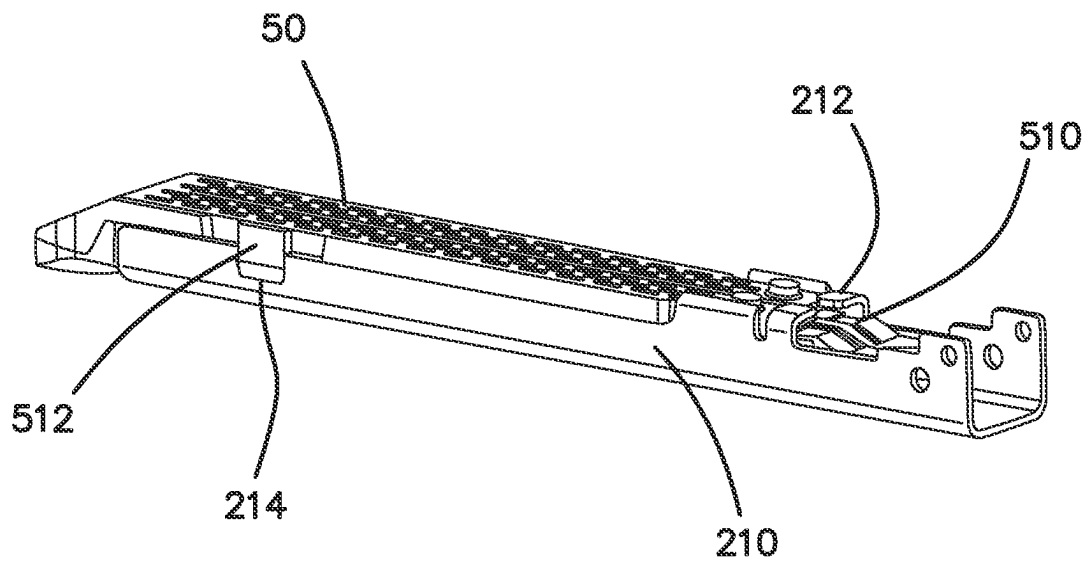


FIG. 13

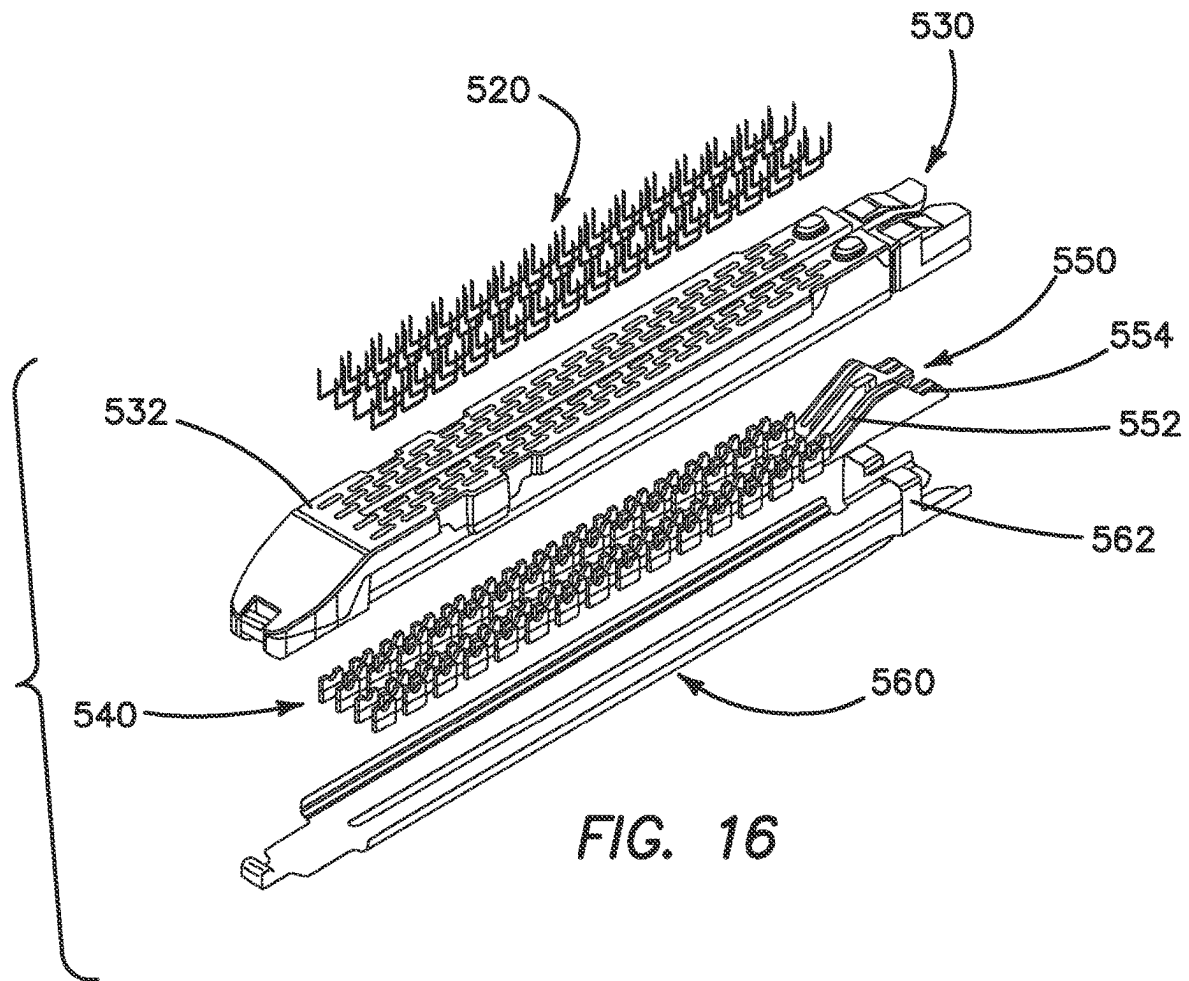


FIG. 16

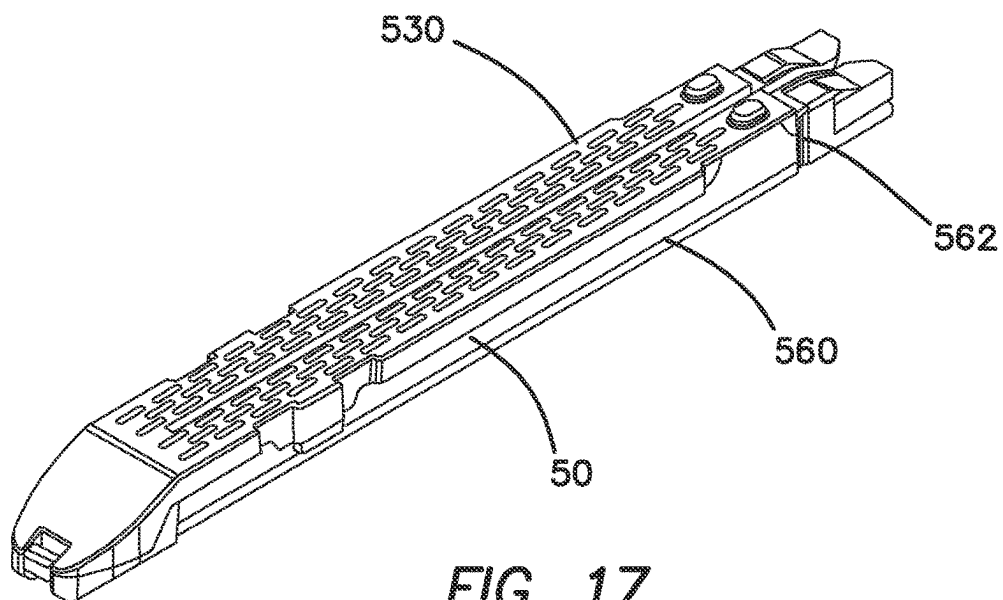


FIG. 17

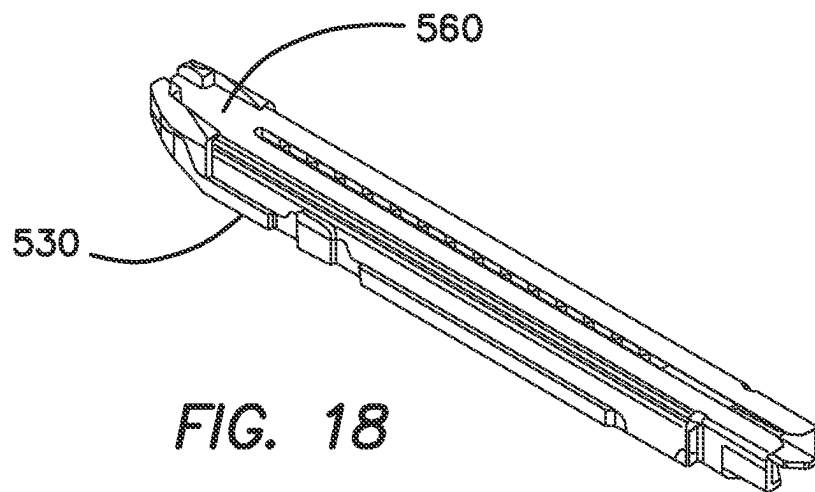


FIG. 18

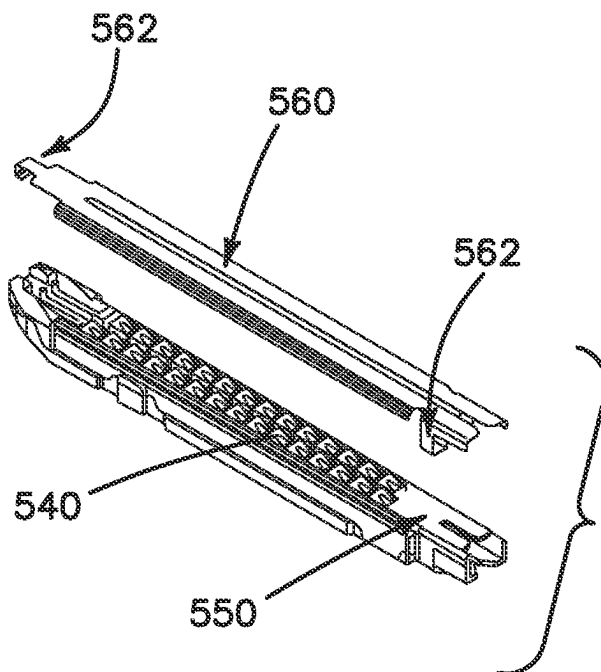


FIG. 19

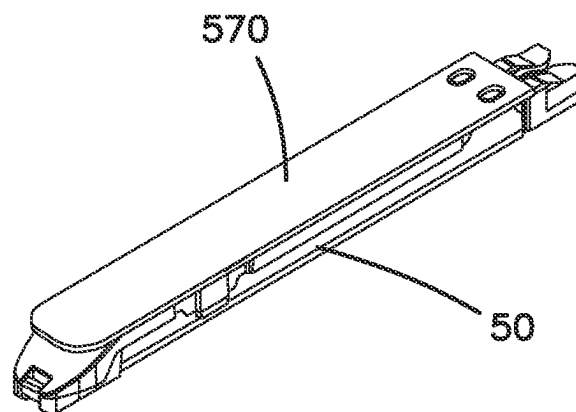


FIG. 20

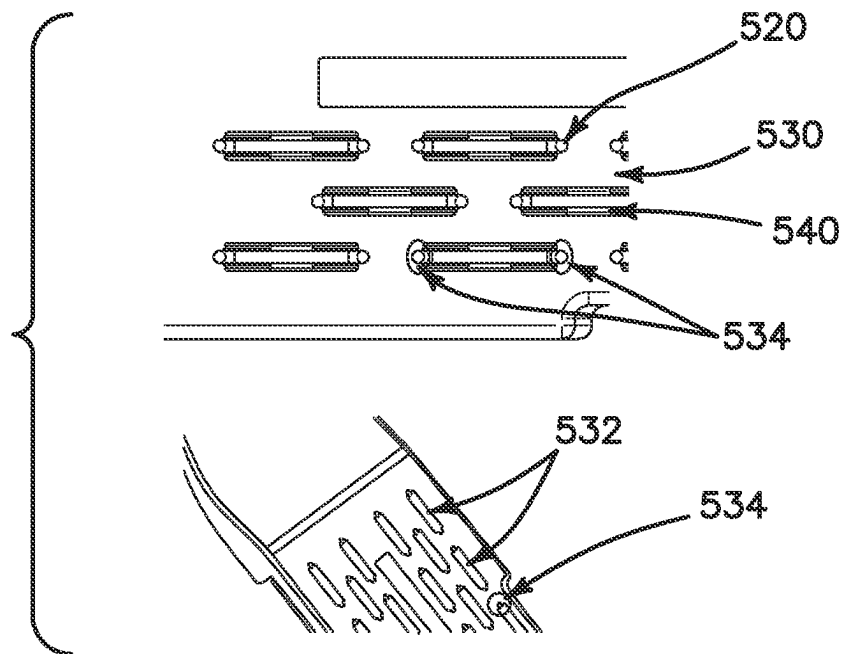


FIG. 21

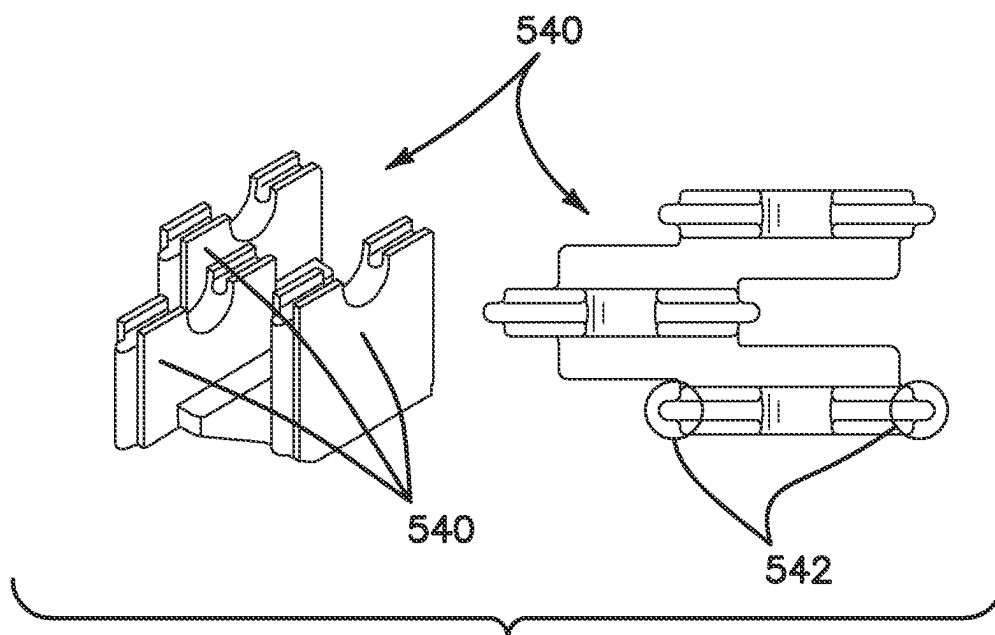


FIG. 22

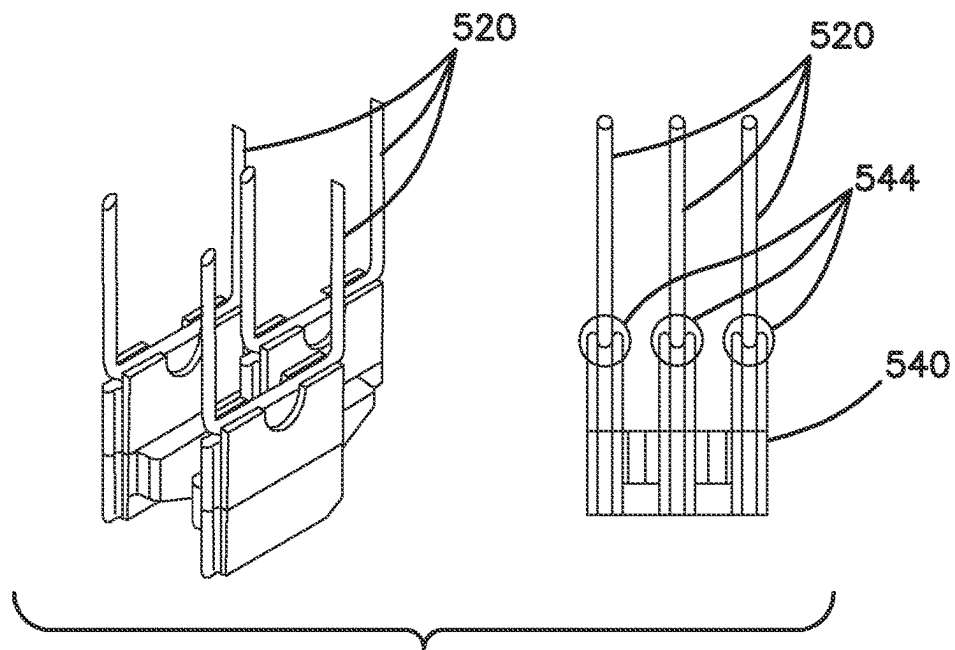


FIG. 23

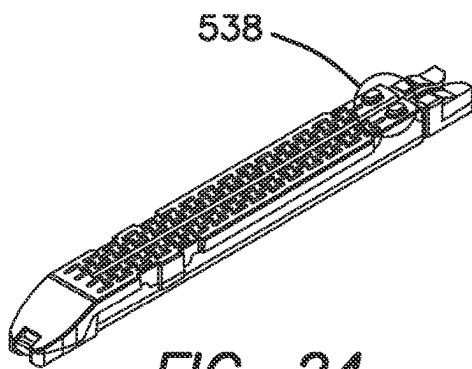


FIG. 24

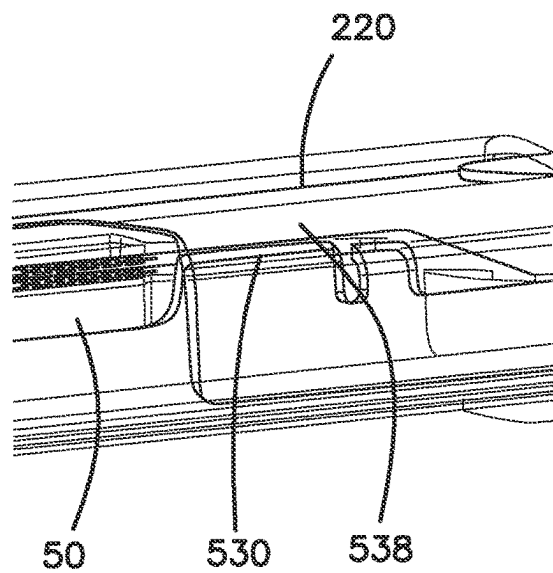


FIG. 25

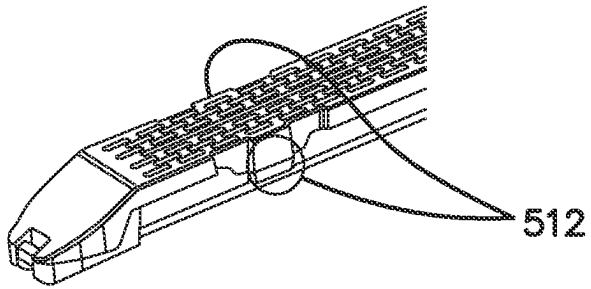


FIG. 26

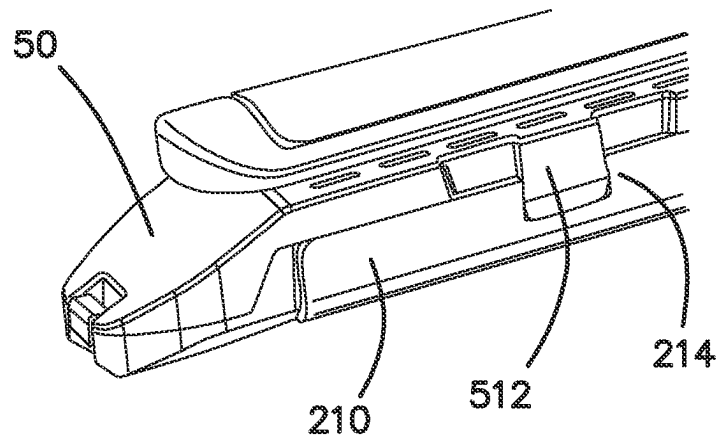


FIG. 27

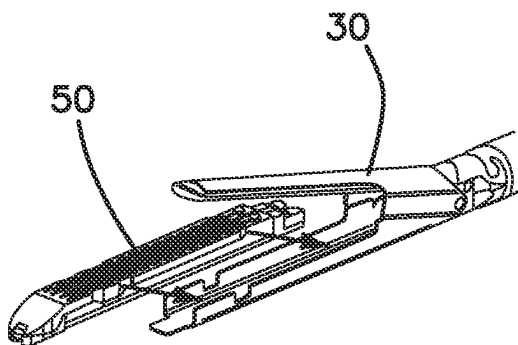


FIG. 28

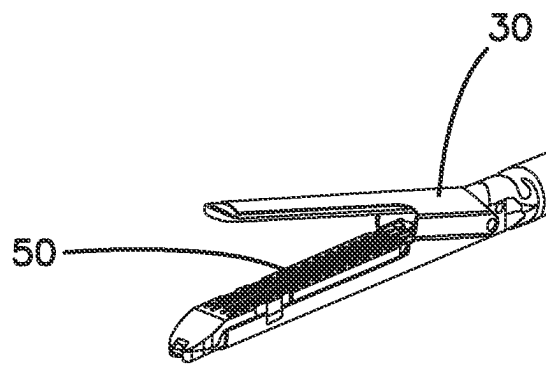


FIG. 29

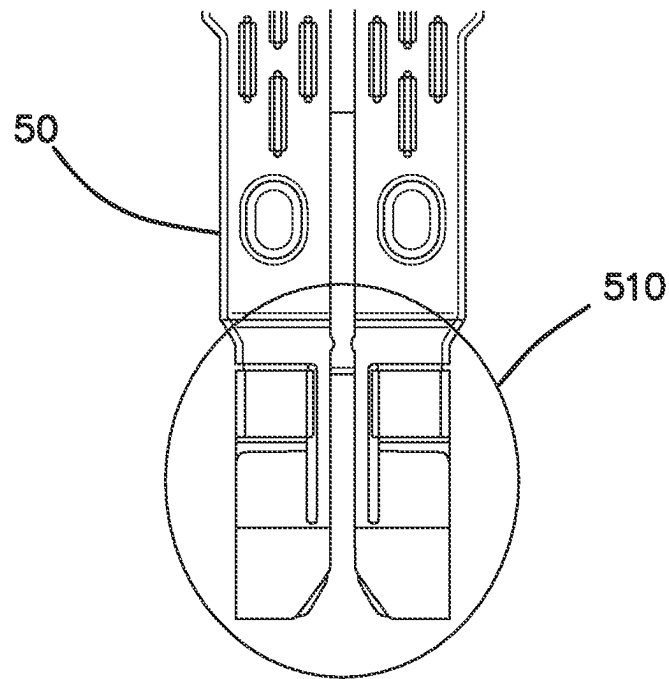


FIG. 30

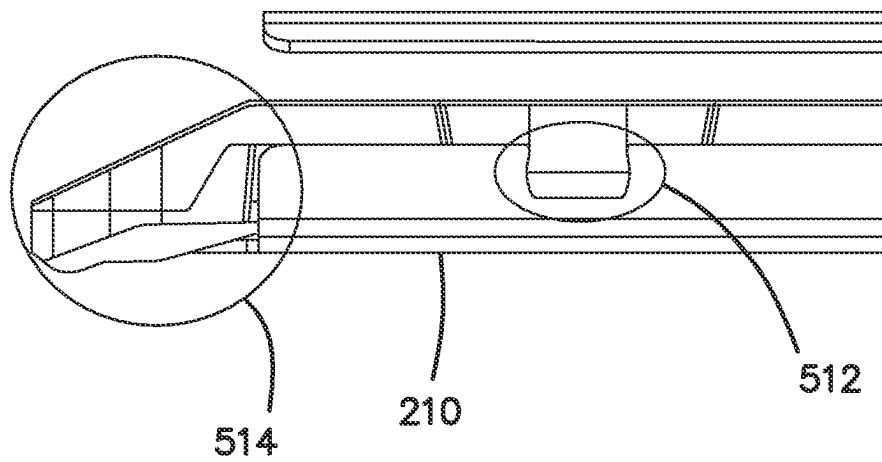


FIG. 31

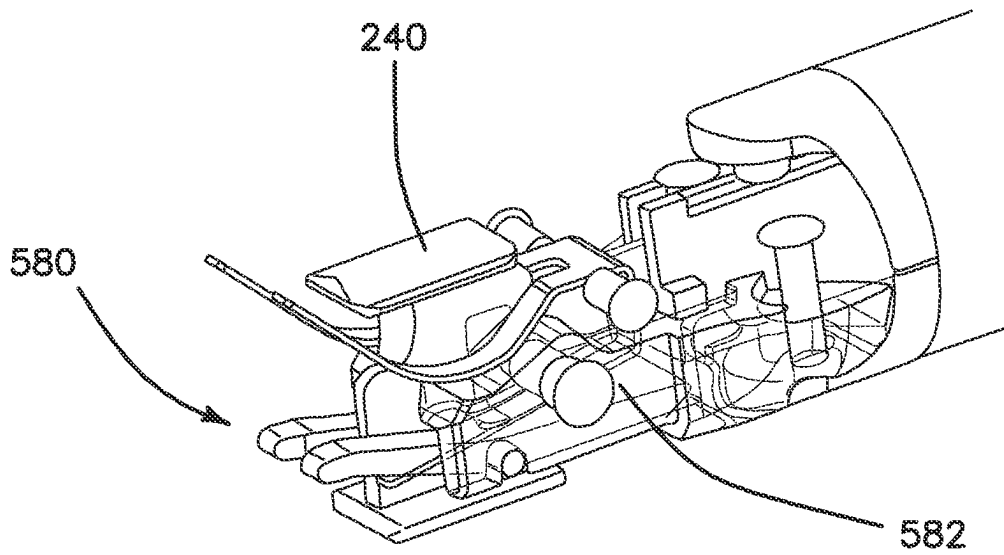


FIG. 32A

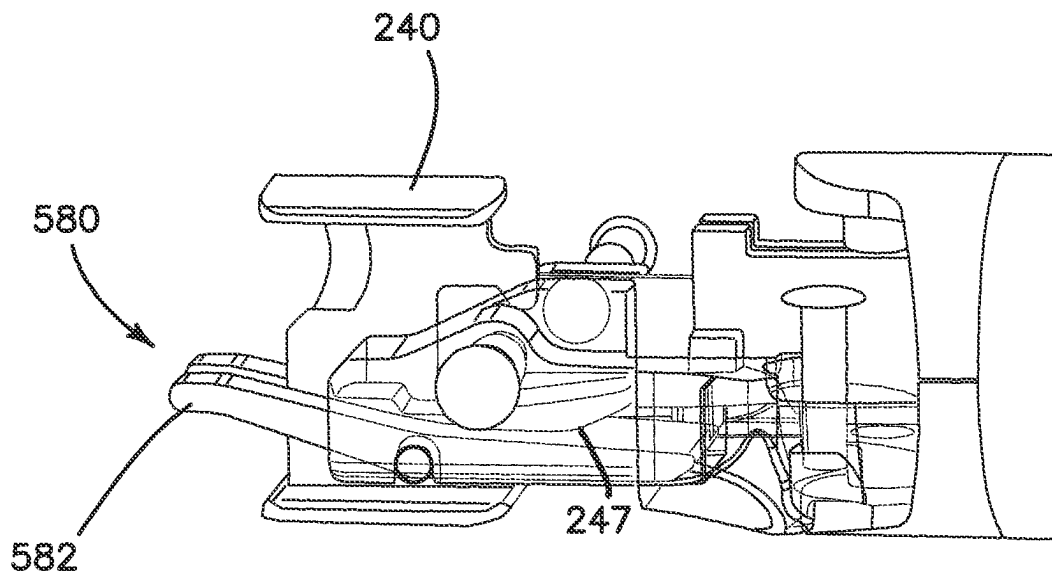


FIG. 32B

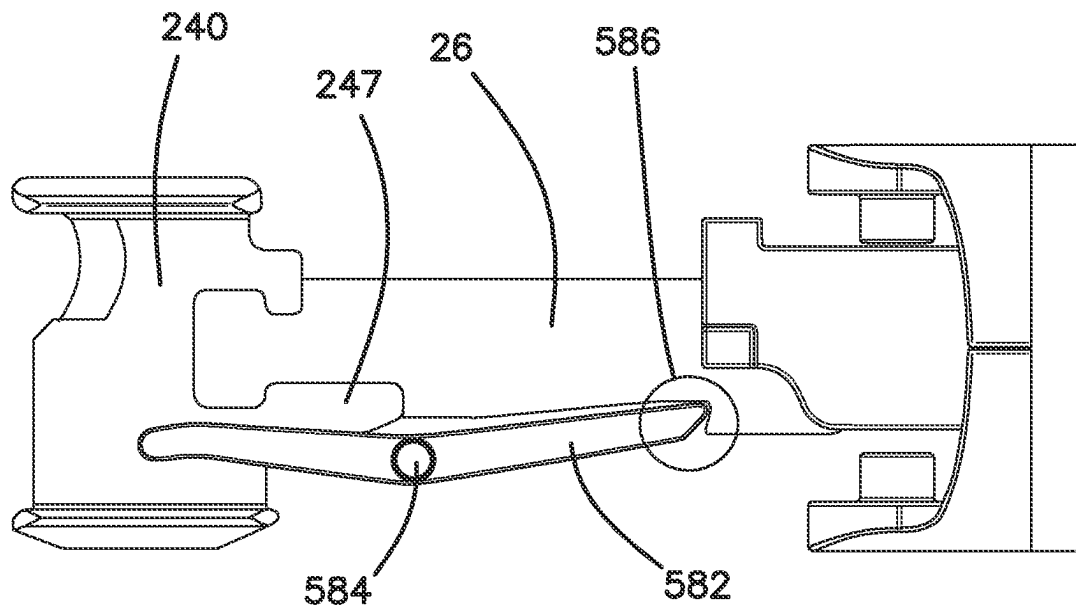


FIG. 33

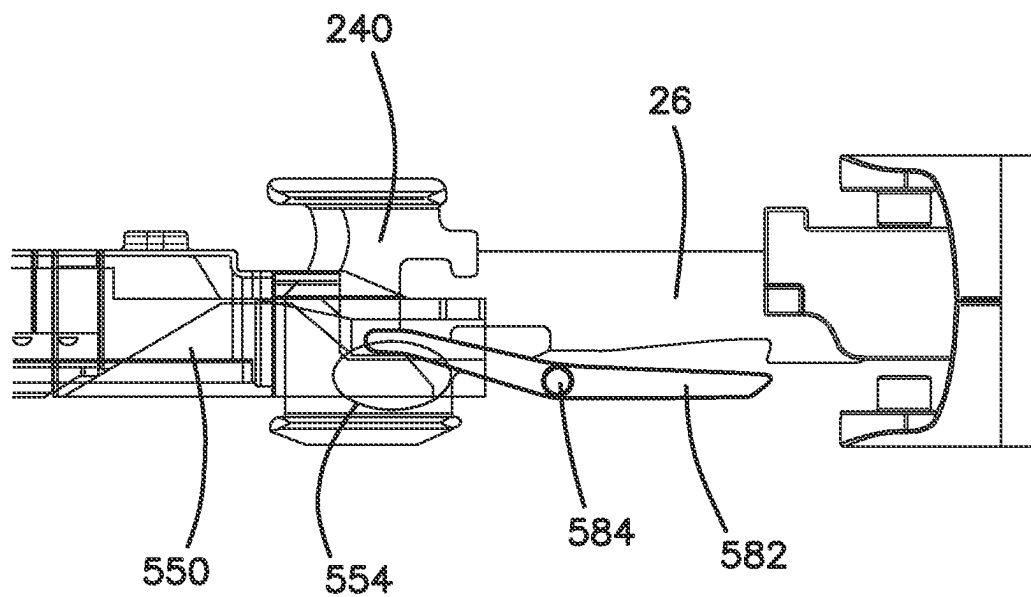


FIG. 34

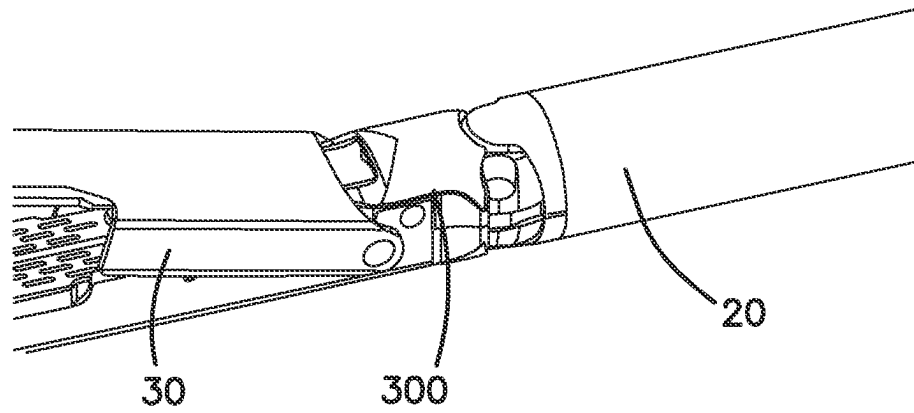


FIG. 35

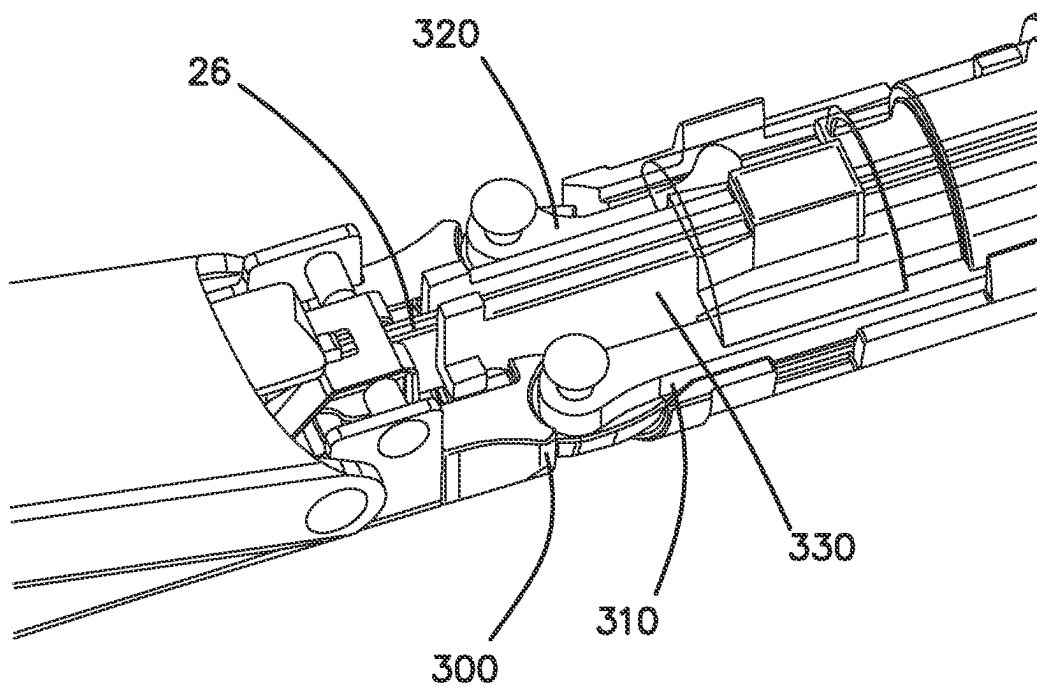
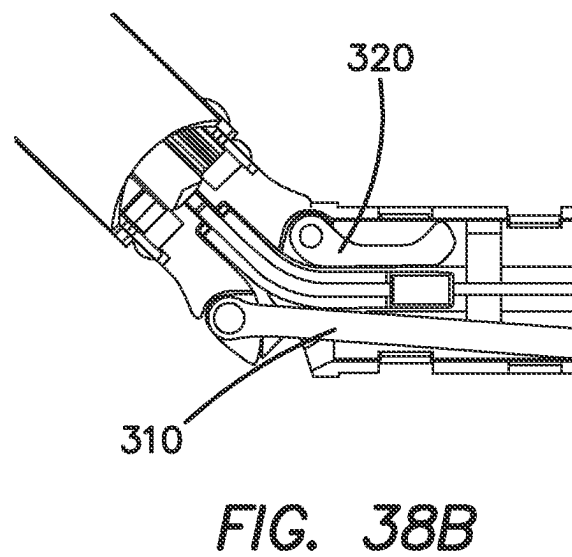
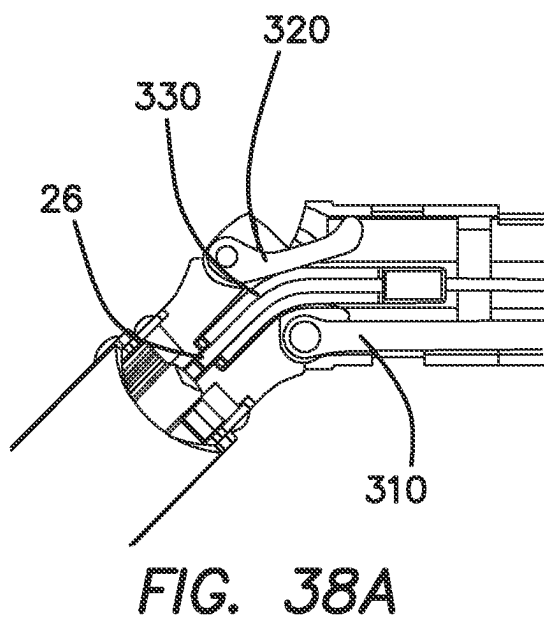
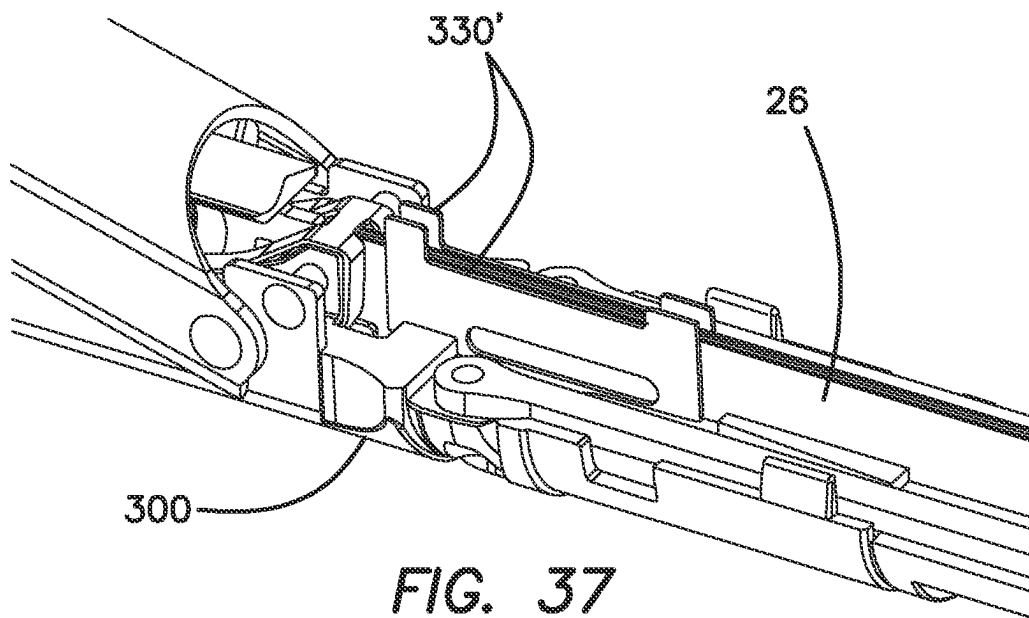
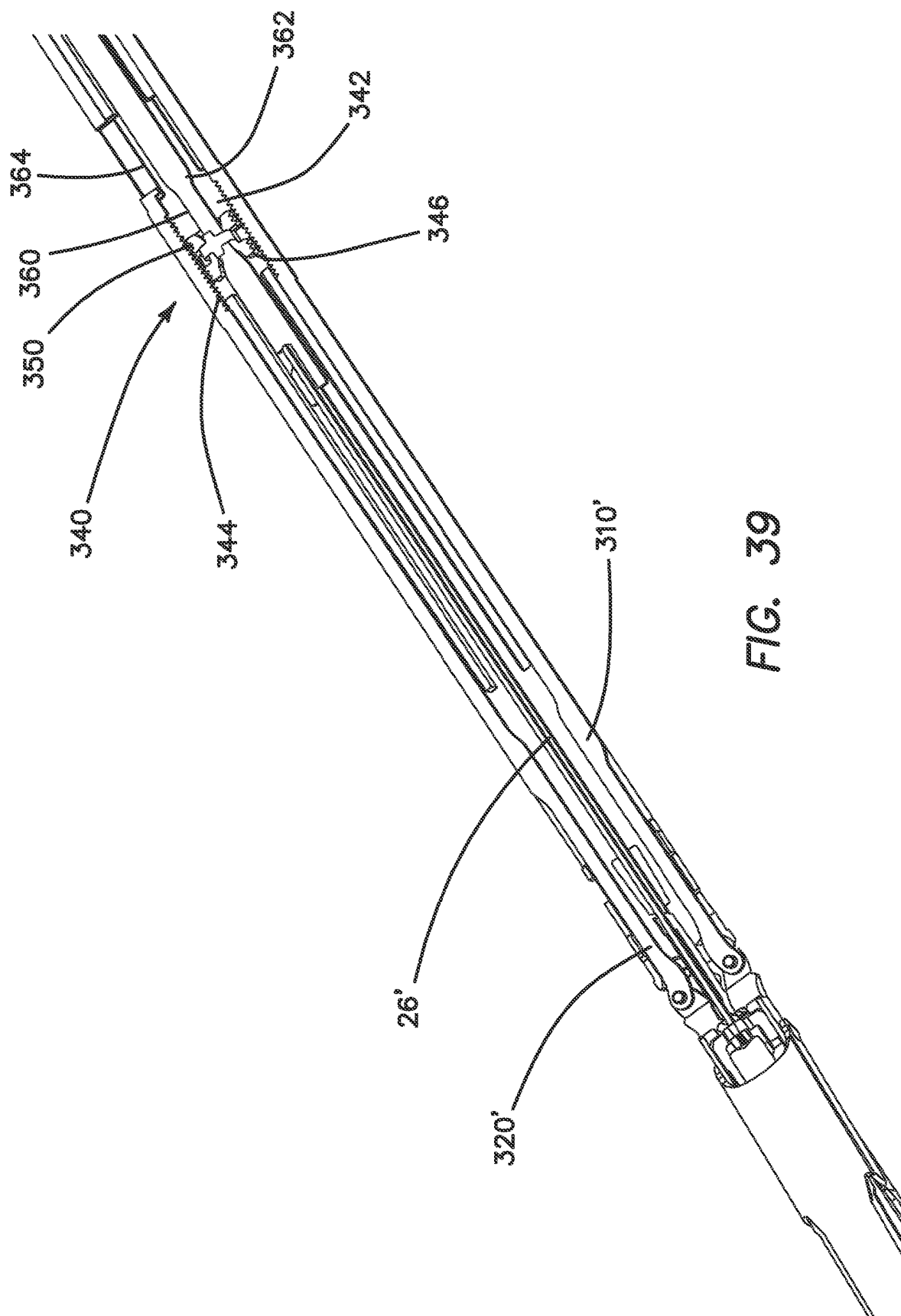
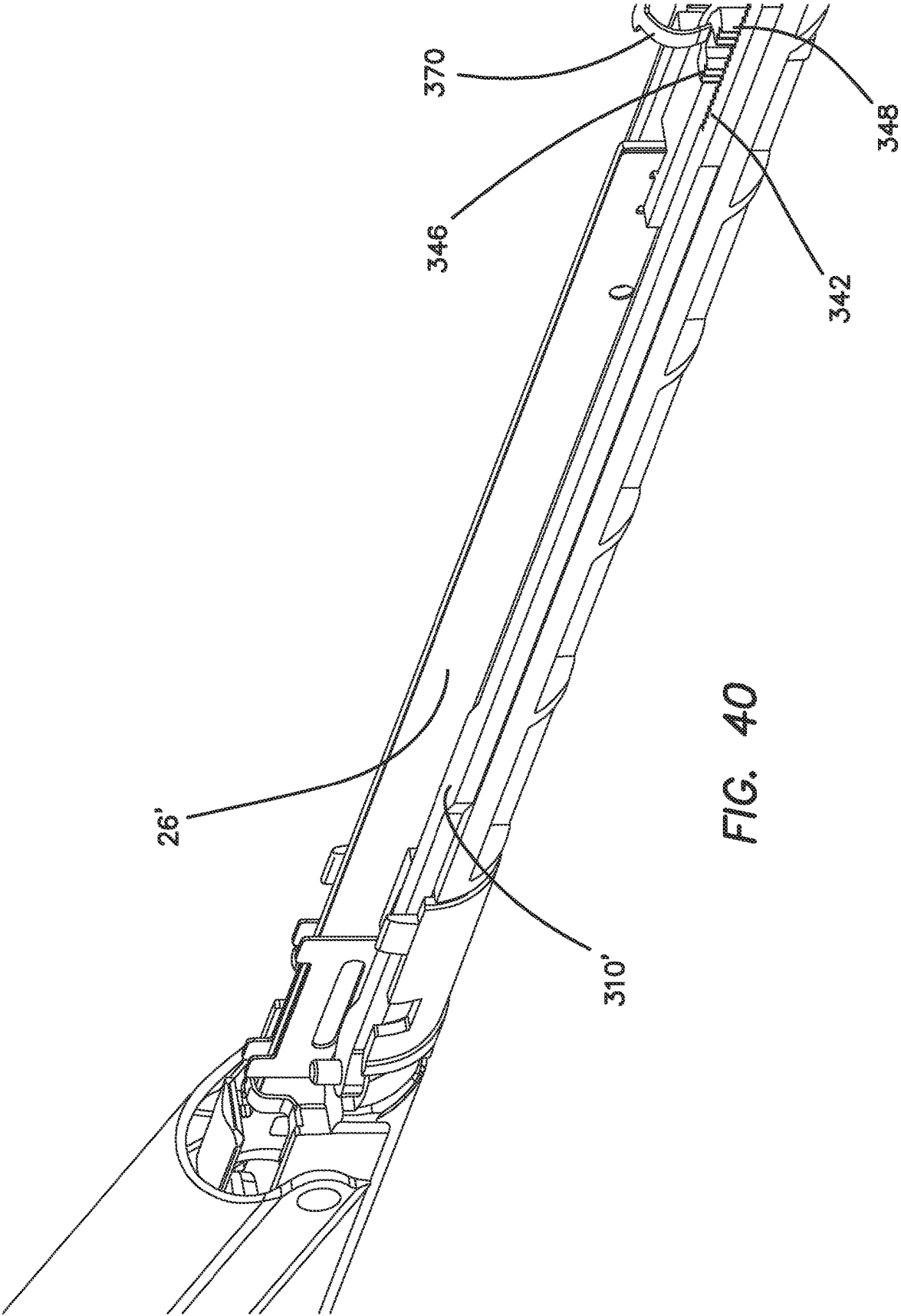
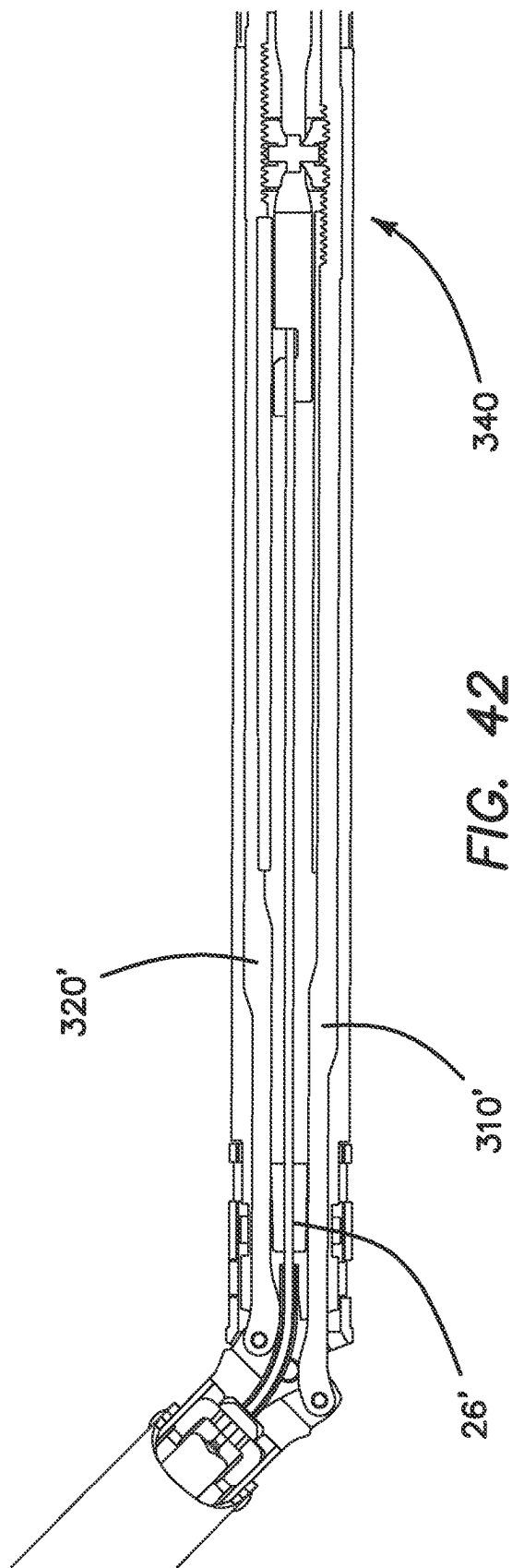
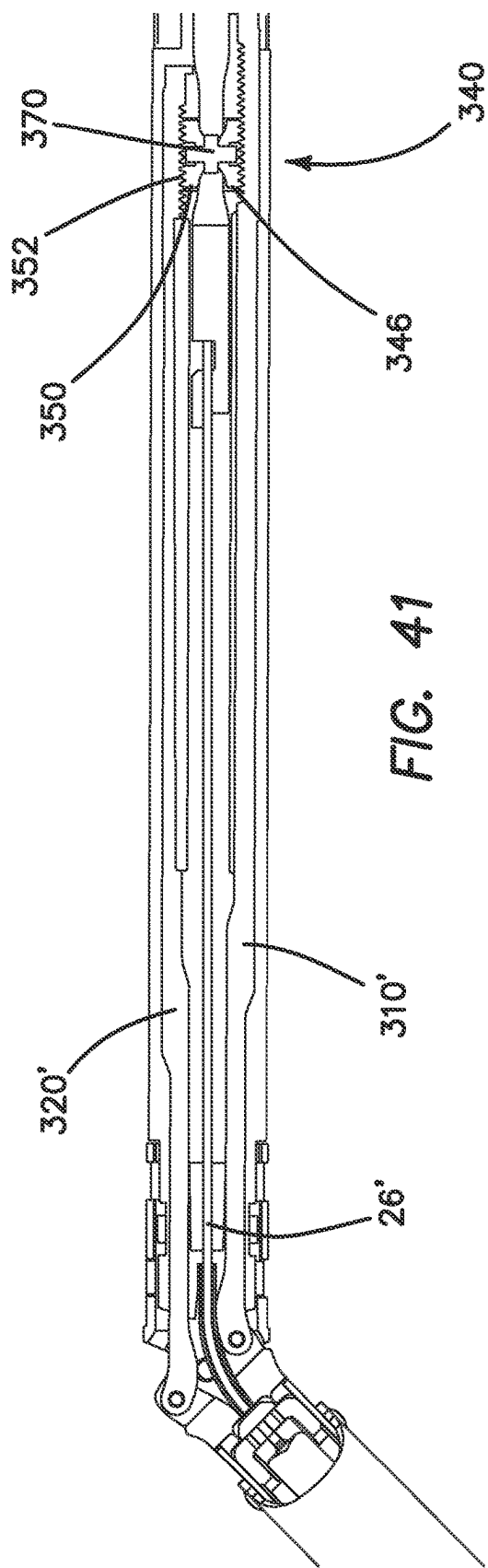


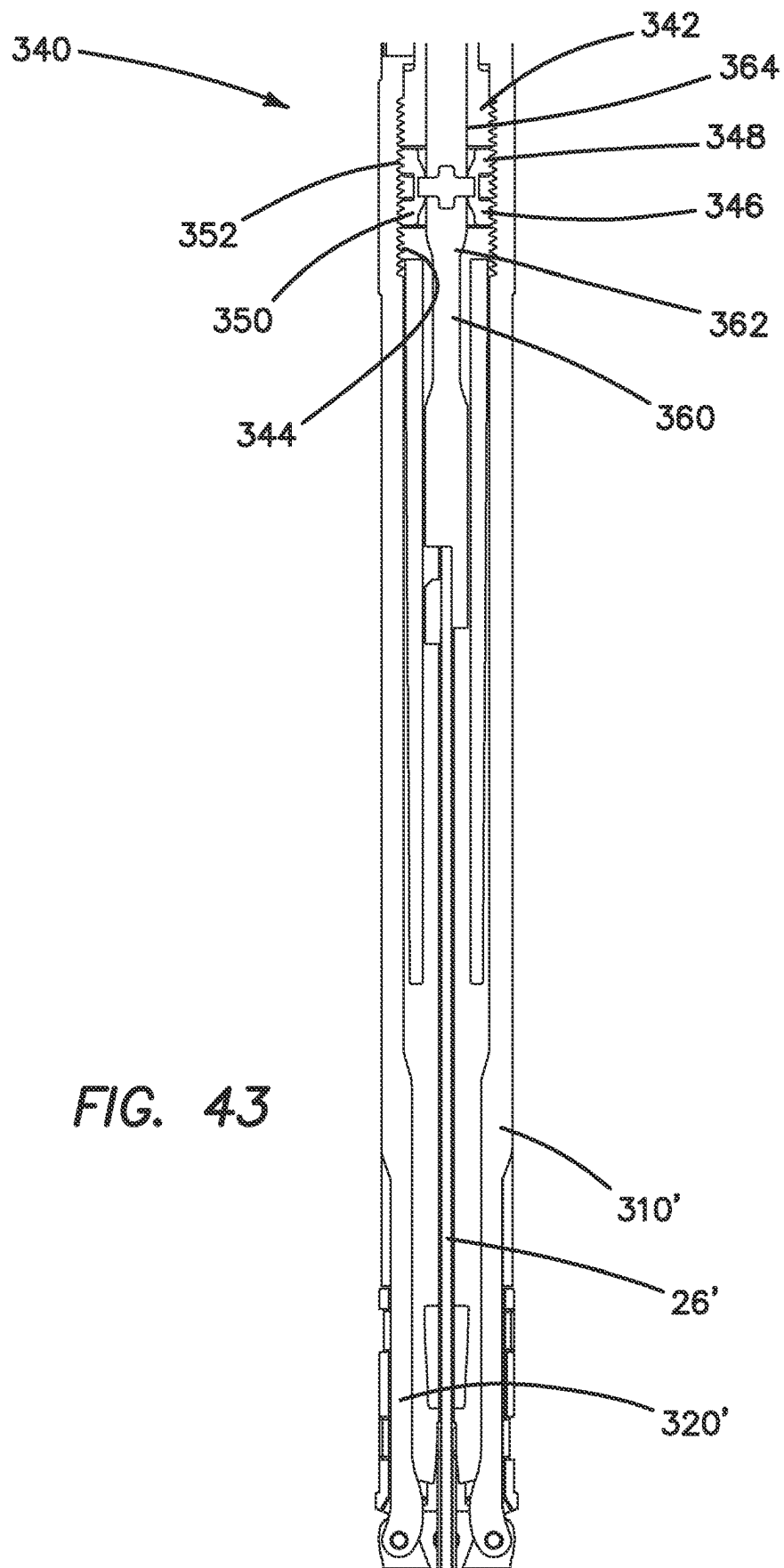
FIG. 36











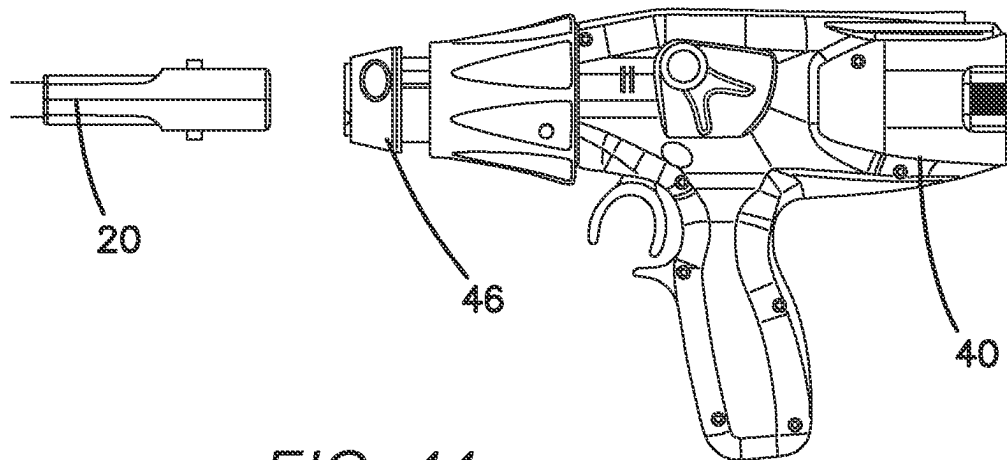


FIG. 44

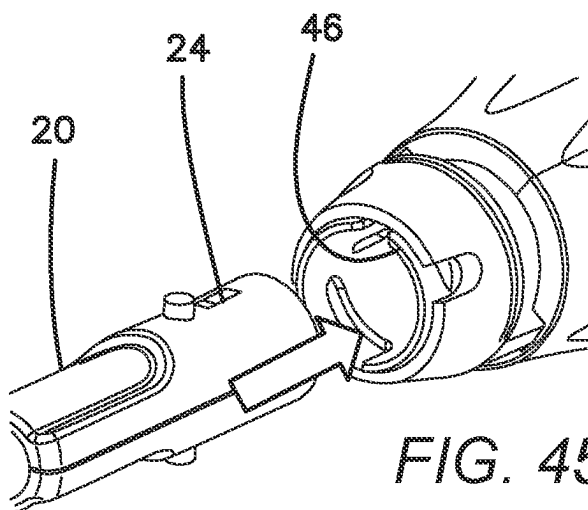


FIG. 45A

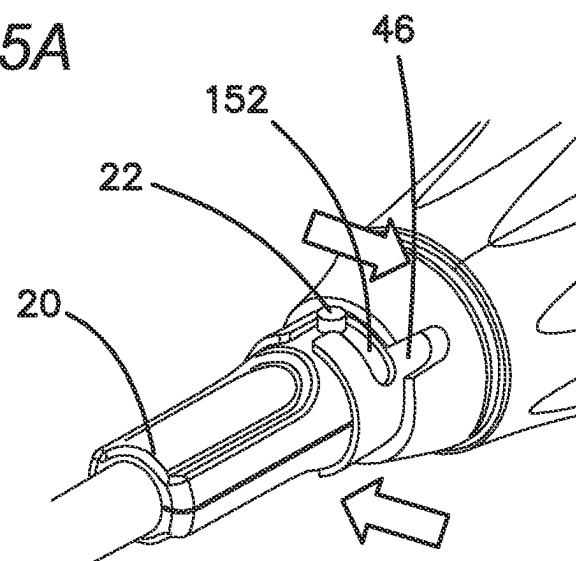


FIG. 45B

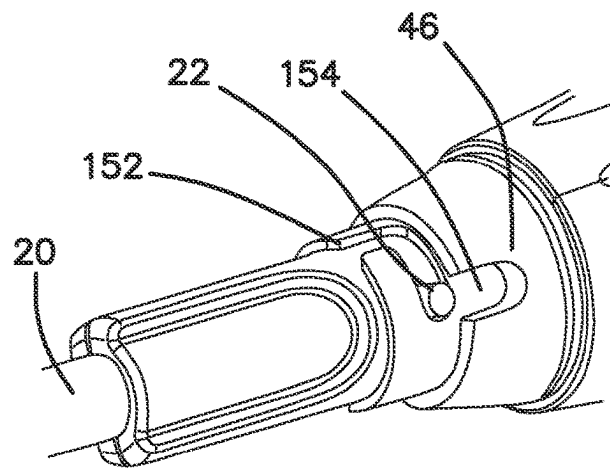


FIG. 45C

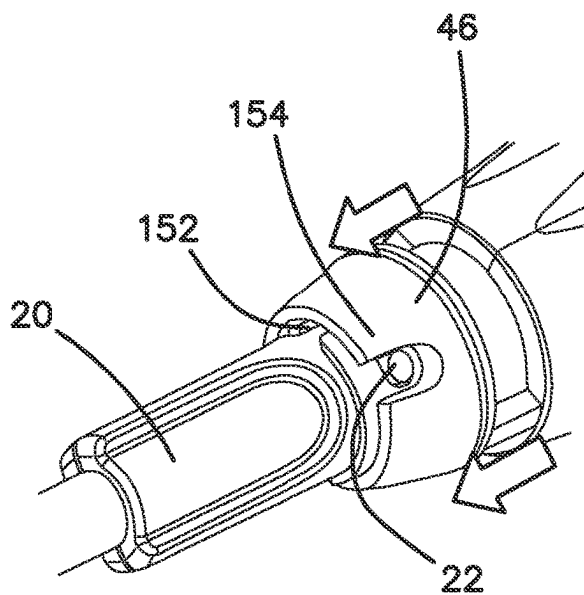


FIG. 45D

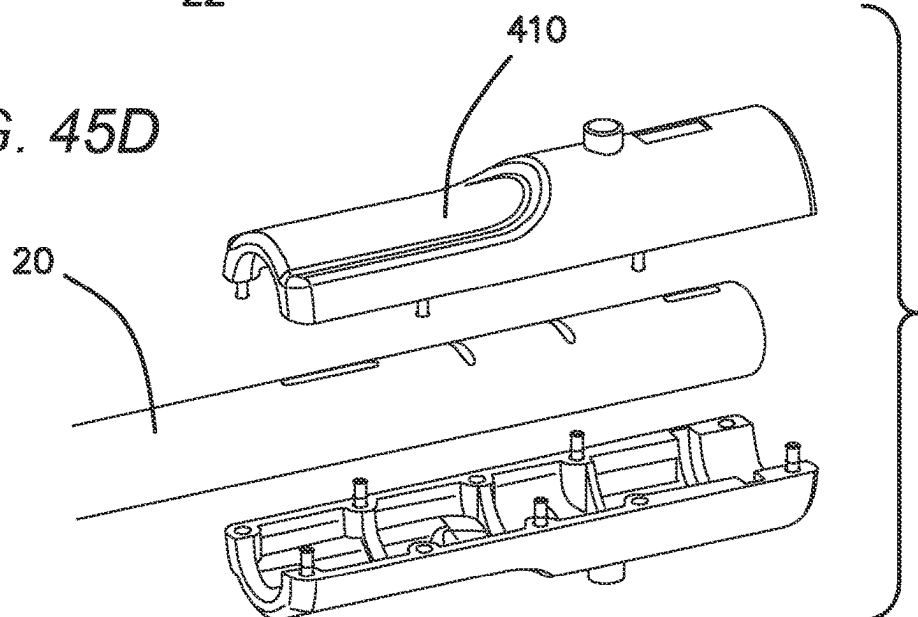


FIG. 52

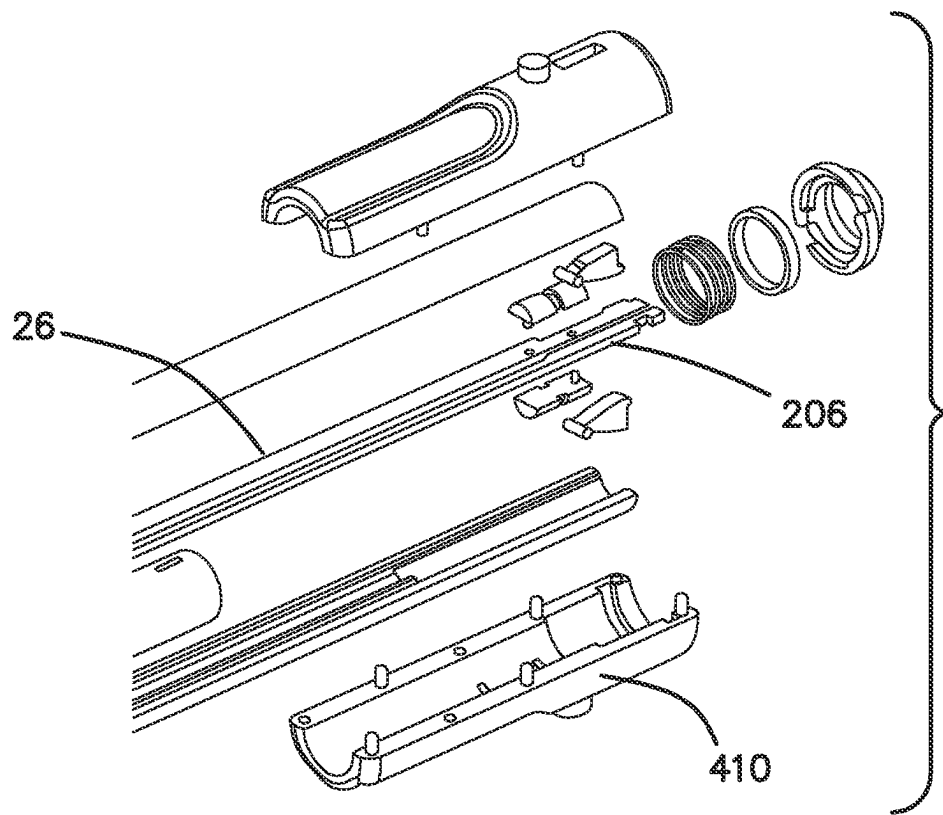


FIG. 46

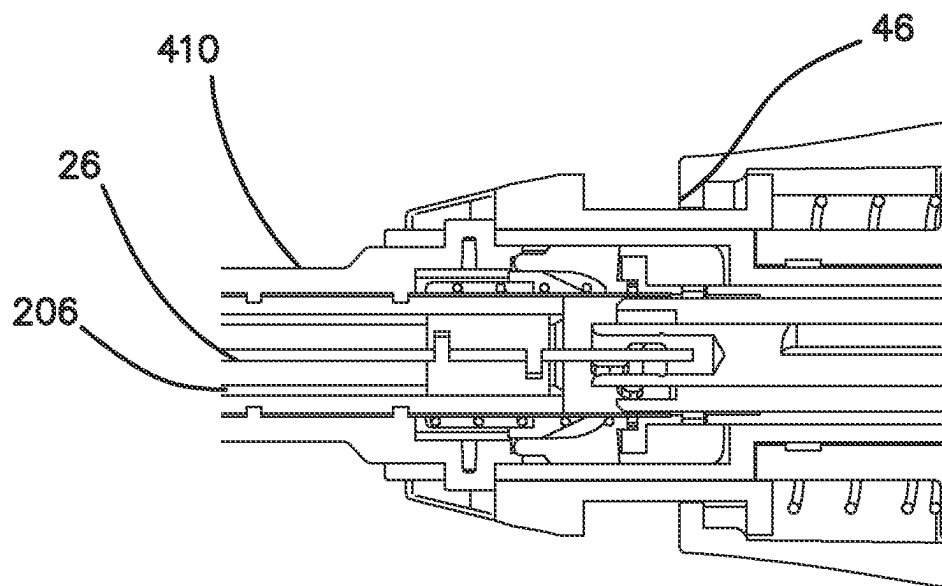


FIG. 47

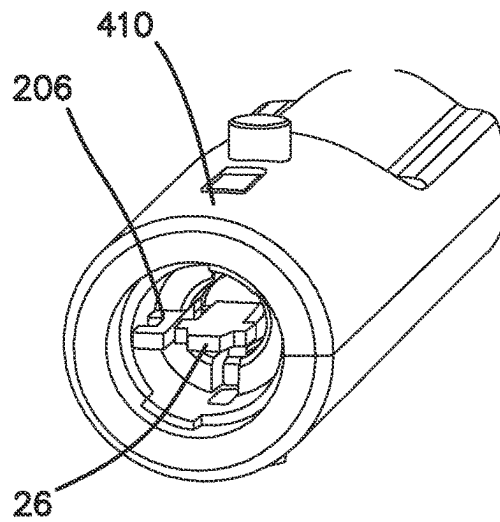


FIG. 48A

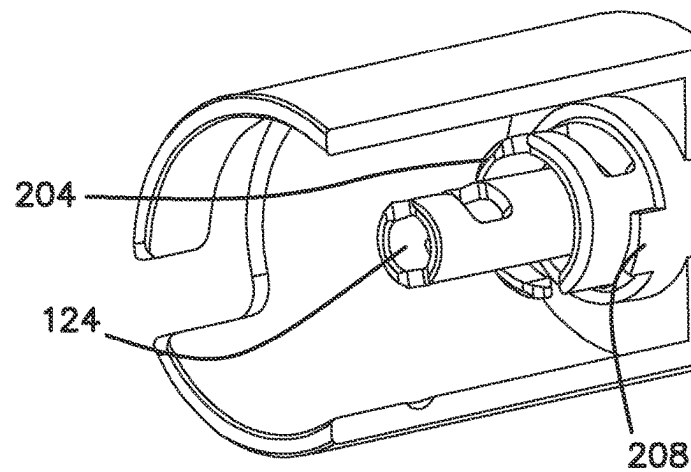


FIG. 48B

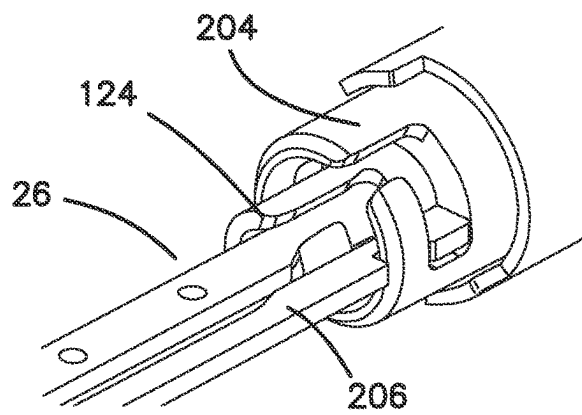


FIG. 49A

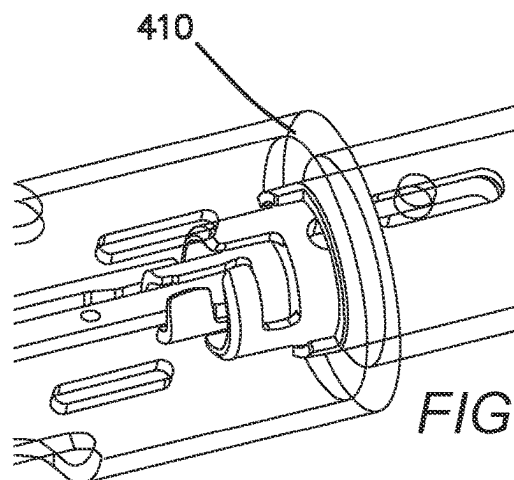


FIG. 49B

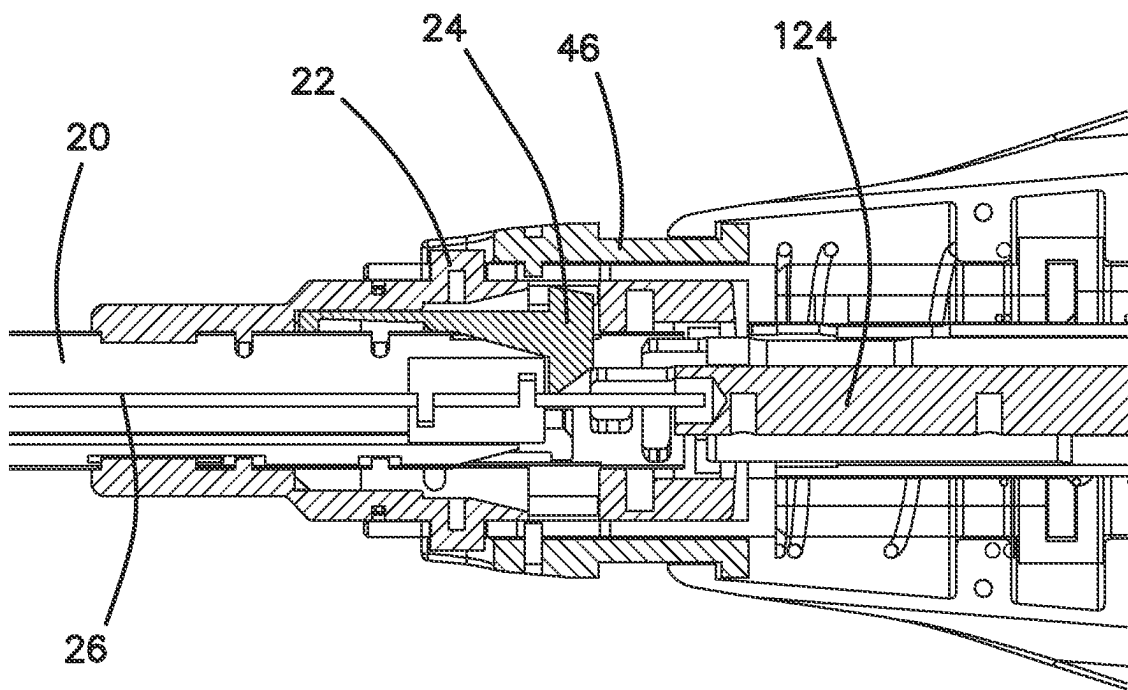


FIG. 50

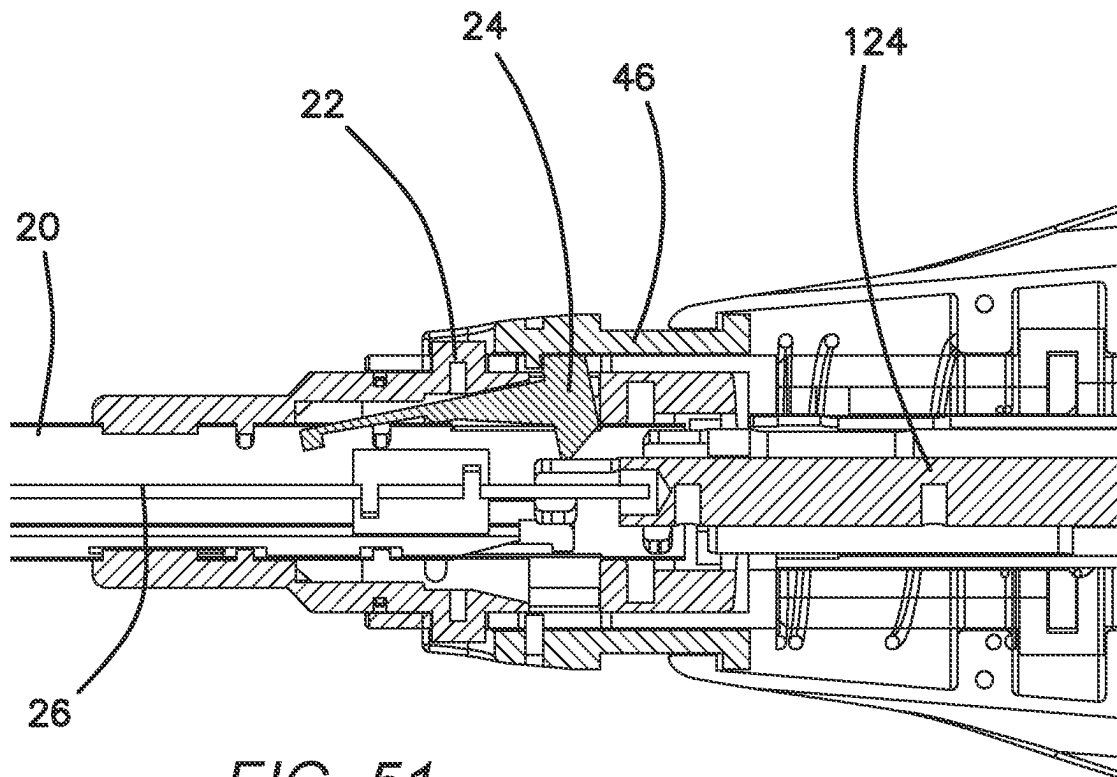


FIG. 51

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**RELOAD SHAFT ASSEMBLY FOR
SURGICAL STAPLER****CROSS REFERENCE TO RELATED
APPLICATION**

This application is a division of U.S. patent application Ser. No. 17/139,518, entitled "RELOAD SHAFT ASSEMBLY FOR SURGICAL STAPLER," filed Dec. 31, 2020, currently pending, which is a division of U.S. patent application Ser. No. 15/486,227, entitled "RELOAD SHAFT ASSEMBLY FOR SURGICAL STAPLER," filed Apr. 12, 2017, now U.S. Pat. No. 10,905,420, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/321,618, entitled "RELOAD SHAFT ASSEMBLY FOR SURGICAL STAPLER," filed Apr. 12, 2016. The above-referenced applications are each incorporated by reference herein in their entireties.

BACKGROUND OF THE INVENTION**Field of the Invention**

The present application relates generally to surgical occlusion instruments and, more particularly, to surgical staplers.

Description of the Related Art

Surgical staplers are used to approximate or clamp tissue and to staple the clamped tissue together. As such, surgical staplers have mechanisms to ensure that tissue is properly positioned and captured and to drive staples through the tissue. As a result, this has produced, for example, multiple triggers and handles in conjunction with complex mechanisms to provide proper stapling of the clamped tissue. With these complex mechanisms, surgical staplers can have increased manufacturing burdens, as well as potential sources for device failure and confusion for the user. Thus, reliable stapling of clamped tissue without complex mechanisms is desired.

SUMMARY OF THE INVENTION

In certain embodiments, a surgical stapler is provided herein. The surgical stapler comprises an elongate shaft, a jaw assembly, and a handle assembly. The elongate shaft has a proximal end and a distal end. The elongate shaft defines a longitudinal axis between the proximal end and the distal end. The jaw assembly is positioned at the distal end of the elongate shaft. The jaw assembly comprises a first jaw, a second jaw, and a plurality of staples. The jaw assembly is selectively positionable in one of a closed configuration, an open configuration, and a firing configuration. The handle assembly is positioned at the proximal end of the elongate shaft.

In certain embodiments, the elongate shaft comprises a jaw assembly at the distal end thereof coupled at an articulation joint. The articulation joint can allow articulation of the jaw assembly about an articulation range. Translation of an articulation member that extends through the elongate shaft articulates the jaw assembly. The elongate shaft further comprises a drive member extending through the elongate shaft. The drive member has a flexible segment extending through the articulation joint. A firing member is coupled to the distal end of the drive member.

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In certain embodiments, the jaw assembly at the distal end of the elongate shaft comprises a reload support and an anvil pivotably coupled to the reload support. A firing member having an I-beam configuration is positioned in the jaw assembly. The jaw assembly can further comprise a lockout mechanism to prevent the firing member from being advanced unless an unfired reload is positioned in the jaw assembly.

In various embodiments, a shaft coupler can be positioned at the proximal end of the shaft. The shaft coupler can be configured to engage a coupler on a handle assembly in a bayonet connection. The bayonet connection simultaneously couples an articulation member, a drive member, and the elongate shaft. The coupler can further comprise a shaft identification mechanism. The coupler can further comprise a lock-in mechanism to retain the shaft assembly in connection with the handle assembly.

In various embodiments, a reload assembly for a surgical stapling system is provided. The reload assembly comprises an elongate shaft, a jaw assembly, a firing member, an actuation beam, and a reload lockout mechanism. The elongate shaft has a proximal end and a distal end. The elongate shaft defines a longitudinal axis extending between the proximal end and the distal end. The jaw assembly is positioned at the distal end of the elongate shaft. The jaw assembly comprises a first jaw, and a second jaw. The first jaw comprises a reload support configured to receive a staple reload. The second jaw is pivotably coupled to the first jaw. The second jaw comprises an anvil surface. The firing member is longitudinally slidable within the jaw assembly. The actuation beam is longitudinally slidable within the elongate shaft. The actuation beam has a proximal end and a distal end. The distal end of the actuation beam is coupled to the firing member. The reload lockout mechanism comprises a lockout lever pivotally coupled to the reload support and pivotable between a locked position preventing distal movement of the actuation beam relative to the elongate shaft and an unlocked position allowing distal movement of the actuation beam relative to the elongate shaft.

In various embodiments, a reload assembly for a surgical stapling system is provided. The reload assembly comprises an elongate shaft, a jaw assembly, an actuation beam, and a shaft coupler. The elongate shaft has a proximal end and a distal end and defines a longitudinal axis extending between the proximal end and the distal end. The jaw assembly is positioned at the distal end of the elongate shaft. The jaw assembly comprises a first jaw, and a second jaw. The first jaw comprises a reload support configured to receive a staple reload. The second jaw is pivotably coupled to the first jaw. The second jaw comprises an anvil surface. The actuation beam is longitudinally slidable within the elongate shaft. The actuation beam has a proximal end and a distal end. The distal end of the actuation beam is coupled to the jaw assembly. The shaft coupler is positioned at the proximal end of the elongate shaft. The shaft coupler comprises a locking member positioned therein. The locking member is radially outwardly advanceable by distal actuation of the proximal end of the actuation beam.

In various embodiments, a reload assembly for a surgical stapling system is provided. The reload assembly comprises an elongate shaft, a jaw assembly, an actuation beam, and a shaft coupler. The elongate shaft has a proximal end and a distal end and defines a longitudinal axis extending between the proximal end and the distal end. The jaw assembly is positioned at the distal end of the elongate shaft. The jaw assembly comprises a first jaw and a second jaw. The first jaw comprises a reload support configured to receive a staple

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reload. The second jaw is pivotably coupled to the first jaw. The second jaw comprises an anvil surface. The actuation beam is longitudinally slidable within the elongate shaft. The actuation beam has a proximal end and a distal end. The distal end of the actuation beam is coupled to the jaw assembly. The shaft coupler is positioned at the proximal end of the elongate shaft. The shaft coupler is configured to removably couple to a handle assembly. The shaft coupler comprises a lockout mechanism positioned therein. The lockout mechanism comprises a locking ring and a lockout member. The locking ring is rotatable about the longitudinal axis. The lockout member is radially outwardly advanceable by rotation of the locking ring.

In various embodiments, a reload assembly for a surgical stapling system is provided. The reload assembly comprises an elongate shaft, a jaw assembly, an actuation beam, an articulation link, a support link, and an articulation latching mechanism. The elongate shaft has a proximal end and a distal end and defines a longitudinal axis extending between the proximal end and the distal end. The jaw assembly is articulably coupled to the elongate shaft at the distal end of the elongate shaft. The jaw assembly comprises a first jaw and a second jaw. The first jaw comprises a reload support configured to receive a staple reload. The second jaw is pivotably coupled to the first jaw. The second jaw comprises an anvil surface. The actuation beam is longitudinally slidable within the elongate shaft to actuate the jaw assembly. The actuation beam has a proximal end and a distal end. The articulation link is longitudinally slidable within the elongate shaft to articulate the jaw assembly relative to the elongate shaft. The articulation link has a proximal end positioned adjacent the proximal end of the elongate shaft and a distal end pivotably coupled to the jaw assembly. The support link is longitudinally slidable within the elongate shaft. The support link has a proximal end extending longitudinally to a distal end pivotably coupled to the jaw assembly. The articulation latching mechanism is positioned within the elongate shaft between the proximal end and the distal end. The articulation latching mechanism has an unlatched configuration in which the articulation link and the support link are slidable within the elongate shaft and a latched configuration wherein the articulation latching mechanism engages the articulation link and the support link to prevent longitudinal sliding of the articulation link and the support link.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of an embodiment of surgical stapling system with the jaws in an open configuration;

FIG. 2a is a perspective view of several embodiments of shaft assembly for the surgical stapling system of FIG. 1;

FIG. 2b is a perspective view of several embodiments of jaw assembly for the surgical stapling system of FIG. 1;

FIG. 3 is a perspective view of a jaw assembly at the distal end of the shaft assembly for the surgical stapling system of FIG. 1;

FIG. 4 is a perspective view of the jaw assembly of FIG. 3 with a staple reload;

FIG. 5 is a perspective view of the jaw assembly of FIG. 3 with a staple reload inserted;

FIG. 6A is a top view of an anvil for the jaw assembly of FIG. 3;

FIG. 6B is a top view of an anvil plate for the jaw assembly of FIG. 3;

FIG. 7 is an exploded perspective view of the anvil of the jaw assembly of FIG. 3;

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FIG. 8 is a perspective view of the top jaw of the jaw assembly of FIG. 3 in an initial state and a formed state;

FIG. 9 is a perspective view of the anvil surface of the jaw assembly of FIG. 3;

FIG. 10 schematic diagram of staple recesses in the anvil surface of FIG. 9;

FIG. 11A is a perspective view of the anvil of the jaw assembly of FIG. 3;

FIG. 11B is a top view of the anvil of the jaw assembly of FIG. 3;

FIG. 12 is a perspective view of the reload support of the jaw assembly of FIG. 3 with a reload partially inserted;

FIG. 13 is a perspective view of the reload support of the jaw assembly of FIG. 3 with a reload inserted;

FIG. 14 is a side view of a closure beam of the jaw assembly of FIG. 3;

FIG. 15 is a partial cut-away front view of the closure beam of FIG. 14 with a flange thereof positioned in a channel in the anvil of the jaw assembly of FIG. 3;

FIG. 16 is an exploded perspective view of a reload for use in the staple system of FIG. 1;

FIG. 17 is an upper perspective view of the reload for use in the staple system of FIG. 1;

FIG. 18 is a lower perspective view of the reload for use in the staple system of FIG. 1;

FIG. 19 is an exploded lower perspective view of the reload for use in the staple system of FIG. 1;

FIG. 20 is a perspective view of the reload for use in the staple system of FIG. 1;

FIG. 21 is a top detail view of the reload for use in the staple system of FIG. 1;

FIG. 22 is a perspective view of a staple pusher for the reload of FIG. 16;

FIG. 23 is a perspective view of a staple pusher of the reload of FIG. 16;

FIG. 24 is a perspective view of the reload for use in the staple system of FIG. 1;

FIG. 25 is a partial cut-away view of the jaw assembly of FIG. 3 in a closed configuration;

FIG. 26 is a perspective view of the reload of FIG. 16;

FIG. 27 is a perspective view of the jaw assembly of FIG. 3 in a closed configuration with a reload inserted;

FIG. 28 is a perspective view of the jaw assembly of FIG. 3 with a reload positioned for insertion;

FIG. 29 is a perspective view of the jaw assembly of FIG. 3 with a reload inserted;

FIG. 30 is a top view of the reload of FIG. 16;

FIG. 31 is a side view of the jaw assembly of FIG. 3 with a reload inserted;

FIG. 32A is a perspective view of a reload lockout mechanism of the shaft assembly;

FIG. 32B is a side view of the reload lockout mechanism of the shaft assembly;

FIG. 33 is a side view of the reload lockout mechanism of the shaft assembly in a locked configuration;

FIG. 34 is a side view of the reload lockout mechanism of the shaft assembly in an unlocked configuration;

FIG. 35 is a perspective view of the distal end of the elongate shaft at an articulation joint connection with the jaw assembly of FIG. 3;

FIG. 36 is a partial cut-away perspective view of one embodiment of articulation joint at the distal end of the elongate shaft;

FIG. 37 is a partial cut-away perspective view of one embodiment of articulation joint at the distal end of the elongate shaft;

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FIG. 38A is a partial cut-away top view of the articulation joint of FIG. 36 in an articulated position;

FIG. 38B is a partial cut-away top view of the articulation joint of FIG. 36 in another articulated position;

FIG. 39 is a partial cut-away perspective view of another embodiment of articulation joint at the distal end of the elongate shaft;

FIG. 40 is a partial cut-away perspective view of the embodiment of articulation joint of FIG. 39 at the distal end of the elongate shaft;

FIG. 41 is a partial cut-away top view of the articulation joint of FIG. 39 in an articulated position;

FIG. 42 is a partial cut-away top view of the articulation joint of FIG. 39 in another articulated position;

FIG. 43 is a partial cut-away top view of the articulation joint of FIG. 39 in a latched position;

FIG. 44 is a side view of the proximal end of the shaft assembly positioned adjacent a handle assembly for the stapler system of FIG. 1;

FIGS. 45A-45D are perspective views of a coupling of the proximal end of the shaft assembly to the handle assembly in a stapler system of FIG. 1;

FIG. 46 is an exploded perspective view of the proximal end of the shaft assembly of the stapler system of FIG. 1;

FIG. 47 is a cut-away side view of the proximal end of the shaft assembly positioned adjacent a handle assembly for the stapler system of FIG. 1;

FIGS. 48A-48B are perspective views of a coupling of the proximal end of the shaft assembly to the handle assembly in a stapler system of FIG. 1;

FIGS. 49A-49B are perspective partial cut-away views of a coupling of the proximal end of the shaft assembly to the handle assembly in a stapler system of FIG. 1;

FIG. 50 is a perspective partial cut-away view of the proximal end of the shaft assembly in a stapler system of FIG. 1;

FIG. 51 is a perspective partial cut-away view of the proximal end of the shaft assembly in a stapler system of FIG. 1; and

FIG. 52 is an exploded perspective view of the proximal end of the shaft assembly in a stapler system of FIG. 1.

DETAILED DESCRIPTION OF THE INVENTION

With reference to FIG. 1, an embodiment of surgical stapling system is illustrated. The illustrated embodiment of surgical stapler 10 comprises an elongate shaft 20, a jaw assembly 30, and a handle assembly 40. FIG. 1 illustrates the surgical stapler 10 with the jaw assembly 30 in an open configuration. A staple reload 50 can be positioned in the jaw assembly. While the illustrated surgical stapling system is illustrated with a powered handle, it is contemplated that the elongate shaft 20 and jaw assembly 30 can be interchangeably used in a stapling system including a mechanical stapler handle. For example, it is contemplated that the various embodiments of elongate shaft assembly 20 and jaw assembly 20 described herein can be used interchangeably with either the powered handle assemblies described in U.S. patent application Ser. No. 15/486,008, entitled "SURGICAL STAPLER HAVING A POWERED HANDLE," filed Apr. 12, 2017, currently pending, and the mechanical manually actuated handle assemblies described in U.S. patent application Ser. No. 15/485,620, entitled "SURGICAL STAPLER HAVING ARTICULATION MECHANISM," filed Apr. 12, 2017, currently pending. These applications are incorporated by reference herein in their entireties.

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With continued reference to FIG. 1, the illustrated embodiment of surgical stapler 10 can be sized and configured for use in laparoscopic surgical procedures. For example, the elongate shaft 20 and jaw assembly 30 can be sized and configured to be introduced into a surgical field through an access port or trocar cannula. In some embodiments, the elongate shaft 20 and jaw assembly 30 can be sized and configured to be inserted through a trocar cannula having a relatively small working channel diameter, such as, for example, less than 8 mm. In other embodiments, elongate shaft 20 and jaw assembly 30 can be sized and configured to be inserted through a trocar cannula having a larger working channel diameter, such as, for example, 10 mm, 11 mm, 12 mm, or 15 mm. In other embodiments, it is contemplated that certain aspects of the surgical staplers described herein can be incorporated into a surgical stapling device for use in open surgical procedures.

With continued reference to FIG. 1, as illustrated, the elongate shaft 20 comprises a generally tubular member. The elongate shaft 20 extends from a proximal end to a distal end. The elongate shaft 20 defines a central longitudinal axis, L, of the surgical stapler 10 extending between the proximal end and the distal end.

With reference to FIG. 2a, it is contemplated that the stapling system can include an elongate shaft having a desired length. While the features of the jaw assembly and handle coupling described herein can be substantially similar for each of these shaft assemblies, the shaft bodies can be scalable. For example, a stapling system can include a relatively short elongate shaft 20', a mid-length elongate shaft 20, or a relatively long elongate shaft 20". Each of these shaft lengths can have particular applicability for a subset of patients or procedures. For example, the short elongate shaft 20' can be useful in pediatric procedures, and the long elongate shaft 20" can be useful in bariatric procedures.

With reference to FIG. 2b, it is contemplated that the stapling system can include a jaw assembly having a desired length. While the features of the jaw assembly and articulation joint described herein can be substantially similar for each of these shaft assemblies, the jaw assemblies bodies can be scalable. For example, a stapling system can include a relatively short jaw assembly 30', a mid-length jaw assembly 30, or a relatively long jaw assembly 30". Each of these jaw assemblies can have particular applicability for a subset of patients or procedures. In certain embodiments, it is contemplated that the jaw assembly have a length of approximately 45 mm. In other embodiments, it is contemplated that the jaw assembly have a length of approximately 60 mm.

With continued reference to FIG. 1, in the illustrated embodiment, the jaw assembly 30 is coupled to the elongate shaft 20 at the distal end 24 of the elongate shaft 20. The jaw assembly 30 comprises a first jaw 32 and a second jaw 34 pivotally coupled to the first jaw 32. In the illustrated embodiment, the jaw assembly 30 is articulable with respect to the elongate shaft 20.

With continued reference to FIG. 1, in the illustrated embodiment, the jaw assembly 30 can be actuated from an open configuration (FIG. 1) to a closed configuration to a stapling configuration by an actuation member or beam that is longitudinally slidable within the elongate shaft. In an initial position, the beam can be positioned at the distal end of the elongate shaft 20. With the beam in the initial position, the second jaw 34 is pivoted away from the first jaw 32 such that the jaw assembly 30 is in the open configuration. The actuation beam engages the second jaw 34 upon translation

of the actuation member or beam distally along the longitudinal axis L. Translation of the actuation beam distally from the initial position a first distance can actuate the jaw assembly from the open configuration to the closed configuration. With the jaw assembly 30 in the closed configuration, the actuation beam can be returned proximally the first distance to return the jaw assembly 30 to the open configuration. A distal end of the actuation beam can advance a staple slider configured to deploy staples from the first jaw 32 such that further translation of the actuation beam distally past the first distance deploys the plurality of staples from the reload positioned in the first jaw 32.

With continued reference to FIG. 1, in the illustrated embodiment, the handle assembly is coupled to the elongate shaft 20 at the proximal end of the elongate shaft 20. As illustrated, the handle assembly 40 has a pistol grip configuration with a housing defining a stationary handle 42 and a movable handle 44 or trigger pivotably coupled to the stationary handle 42. It is contemplated that in other embodiments, surgical stapler devices including aspects described herein can have handle assemblies with other configuration such as, for example, scissors-grip configurations, or in-line configurations. As further described in greater detail below, the handle assembly 40 houses an actuation mechanism configured to selectively advance an actuation shaft responsive to movement of the movable handle 44.

With reference to FIG. 3, an embodiment of jaw assembly at the distal end of the shaft assembly 20 is illustrated. In the illustrated embodiment, the jaw assembly comprises a reload support 210 articulably coupled to the distal end of the shaft assembly 20 at an articulation joint 230. An anvil 220 is pivotably coupled to the reload support 210 and defines a top jaw of the jaw assembly 30. A firing member 240 can slide within the jaw assembly to initially close the anvil 220 relative to the reload support 210, then fire staples from a reload. In some embodiments, the firing member 240 has an I-beam configuration with a vertical beam 242 spanning between two horizontally-protruding flanges 244, 246. Advantageously, with an I-beam configuration, one horizontal flange 244 can engage a channel in the anvil 220 and the other flange 246 can engage a channel in the reload or reload support to close the jaw assembly then maintain a desired closed spacing of the jaw assembly when the firing member is advanced distally. In some embodiments, the firing member 240 can comprise a cutting blade 248 formed on or mounted to the vertical beam in an I-beam configuration. This cutting blade can separate tissue as staples are fired to form staple lines on both sides of the separated tissue.

With reference to FIGS. 4 and 5, the reload support 210 can be sized to receive and retain a disposable reload 50. The reload 50 can be lowered and moved proximally into the reload support 210 until mating features on the reload engage corresponding features on the reload support 210.

With reference to FIGS. 6A, 6B, 7, and 8, various aspects of the anvil 220 of the jaw assembly 30 are illustrated. In certain embodiments, the anvil 220 comprises an anvil plate 222 coupled to a top surface 224. The anvil plate can comprise a longitudinal channel 225 formed therein in which a horizontal flange of the firing member rides and a longitudinal slot 227 formed through the longitudinal channel 225 in which the vertical beam of the firing member rides. The top surface 224 can be formed of a sheet of material that is subsequently formed to overly the anvil plate. (FIG. 8 illustrates the flat sheet 224' and shaped top

surface 224). Advantageously, the addition of the top surface 224 to the anvil plate 222 enhances the strength of the anvil of the jaw assembly.

With reference to FIGS. 9 and 10, various aspects of the anvil plate 222 of the jaw assembly 30 are illustrated. The anvil plate comprises a plurality of staple forming pockets 223 thereon. In the illustrated embodiment, the staple forming pockets 223 are positioned in two arrays of three rows with the arrays positioned on either side of the slot for the firing member. Thus, the stapler can form two sets of three linear rows of staples with the sets separated by divided tissue. In other embodiments, it is contemplated that the anvil can include staple forming pockets configured to form other numbers and configurations of staples. The staple forming pockets have a tapered configuration with a relatively large staple entry side narrowing to a relatively small staple formation side. Advantageously, this tapered configuration can guide staples to complete formation and reduce the incidence of poorly formed staples. Adjacent rows of staples can be longitudinally offset from one another such that the relatively wide entry sides of all of the rows are offset from one another to reduce the overall width of the sets of staple rows.

With reference to FIGS. 11a and 11b, in certain embodiments of anvil 220, the top surface 224 can be coupled to the anvil plate 222 by a welding operation along a weld line 226. Advantageously, this closed anvil formed by the welding operation covers the channel for the firing member.

With reference to FIGS. 12 and 13, insertion of a reload 50 in the reload support 210 is illustrated. The reload support can comprise proximal jaw tabs 212 that protrude radially inwardly from side walls of the reload support 210 adjacent the proximal end thereof. The reload can comprise a relatively short, tapered proximal deck 510 sized to be positioned under and retained by the proximal jaw tabs. Moreover, the reload 50 can include retention tabs 512 protruding laterally outwardly adjacent a distal end thereof. The reload support 210 can comprise a corresponding pair of retention recesses 214 sized and configured to receive the retention tabs when the reload is positioned in the reload support.

With reference to FIGS. 14 and 15, an embodiment of firing member 240 having an I-beam configuration is illustrated. In the illustrated embodiment, The firing member comprises a vertical beam 242 having a cutting blade formed therein at a leading edge. The cutting blade comprises a curved cutting blade 248. A trailing edge of the firing member 240 comprises a drive member interface 245 such as a cutout or protrusion to allow the firing member to be securely coupled with the drive member extending through the elongate shaft. The trailing edge of the firing member 240 can further comprise a lockout interface 247, such as a proximally extending 'tail' that can position a reload lockout in an unlocked configuration when the firing member is in a proximal position. The firing member further comprises an upper horizontal flange 244 configured to ride in the channel 225 of the anvil and a lower horizontal flange 246 configured to engage the reload or reload support. As illustrated in FIG. 15, although the firing member has a general I-beam configuration, in some embodiments the horizontal flanges are curved or tapered such to conform with a shape of the channel 225 in the anvil. In some embodiments, the firing member 240 can further be configured to reduce friction during a firing sequence such as by surface finishing operations, addition of a film lubricant, or deposition of a low-friction surface on the firing member, channel, or both.

With reference to FIGS. 16-19, an embodiment of reload 50 for use in the stapling system is illustrated. The reload 50

comprises a plurality of staples **520** positioned in a corresponding plurality of staple pockets **532** formed in a cartridge **530**. The staple pockets **532** are arranged in two sets of three rows each with each set separated by a slot formed through the cartridge **530**. The staples **520** rest in a plurality of staple pushers **540** underlying the staple pockets **532**. A slider **550** having a ramp **552** corresponding to each row of staple pusher **540** and a lockout tail **554** is positioned at the proximal end of the reload. The slider **550** is longitudinally slidable within the reload responsive to movement of the firing member. A jacket **560** underlies the cartridge and maintains the staples and staple pushers in the staple pockets. The jacket can have protruding hooks **562** to engage the cartridge.

With reference to FIG. **20**, in some embodiments, the reload **50** can include a shipping cover **570** covering an upper surface of the cartridge. Advantageously, the shipping cover **570** can prevent one or more of the staples from becoming dislodged from or misaligned within the staple pockets before the reload is used. The shipping cover **570** is removed before the reload **50** is positioned in the reload support.

With reference to FIGS. **21-23**, in some embodiments, the reload **50** can include certain staple alignment and retention features. For example the staple pockets **532** formed in the cartridge **530** can include staple guides **534** at ends thereof to receive legs of the staples **520** positioned therein. The staple pushers **540** can additionally include nubs **542** sized and configured to ride in the staple guides **534**. As illustrated, in certain embodiments, the staple pushers **540** can be formed in groups of three such that one staple pusher **540** can push a single staple in each of three adjacent rows of staples. Moreover, an upper surface **544** of each of the staple pushers **540** can include a staple saddle configuration to relatively securely receive a staple. Secure positioning of the staples **520** in the staple pushers **540** and engagement of the staple legs and nubs **542** of the staple pushers with the staple guides can advantageously reduce the incidence of misaligned or malformed staples.

With reference to FIGS. **24-31**, in various embodiments, the reload **50** and jaw assembly can be configured to be securely coupled to one another to align the staple pockets on the reload **50** with the staple forming pockets on the anvil and maintain the position of the reload **50** in the jaw assembly during staple firing. The reload **50** can include upwardly protruding bosses **538** at a proximal end thereof (FIG. **24**) that define a tissue gap between the anvil **220** and an upper surface of the cartridge **530** of the reload **50** with the jaw assembly in a closed configuration (FIG. **25**). Moreover, the retention tabs **512** formed adjacent the distal end of the reload (FIG. **26**) are positioned within recesses **214** of the reload support **210** and prevent the reload from shifting distally during a firing operation. Thus, the reload **50** can be rapidly and securely coupled to the reload support **210** (FIGS. **28-29**). Additionally, a proximal end of the cartridge **530** can taper to a reduced height to further facilitate placement on the reload support (FIG. **30**). Furthermore, the cartridge can be configured with a lowered distal end **514** having a profile protruding below the reload support (FIG. **31**). This lowered profile ensures secure engagement of the reload with the reload support.

With reference to FIGS. **32A**, **32B**, and **33-34**, in certain embodiments, the jaw assembly can comprise a reload lockout mechanism **580**. The reload lockout mechanism **580** can prevent advancement of the firing member if no reload is positioned within the jaw assembly or if an empty reload is positioned within the jaw assembly. The reload lockout

mechanism **580** includes a lockout lever **582** pivotally coupled to the reload support. An axis defined by the pivot extends generally transverse to the longitudinal axis of the elongate shaft. With the firing member **240** fully retracted such that the jaw assembly is in an open configuration, a tail **247** extending proximally from the firing member **240** maintains the lockout lever **582** pivoted to the unlocked position. In the illustrated embodiment, a proximal portion of the lockout lever **582** proximal the pivot is forked or bifurcated to receive the firing member **240** therein such that the tail **247** can act on a surface of the lockout lever **582** distal the pivot. If no reload is inserted, an attempt to advance the firing member **240** will allow the lockout lever to pivot about a pivot point **584** from the unlocked position to the locked position as the tail **247** of the firing member is advanced distally along the lockout lever. (FIG. **33**). With the lockout lever **582** in the locked position, a proximal, locking end **586** of the lockout lever interferes with a lock recess on the drive member **26**, preventing further distal movement of the drive member.

With continued reference to FIGS. **32A**, **32B**, and **33-34**, if an unfired reload is inserted into the reload support (FIG. **34**), a tail **554** extending proximally from the slider **550** engages a distal end of the lockout lever **582**. As illustrated, the tail **554** acts on a lower surface of a distal portion of the lockout lever **582** distal the pivot point. This engagement of the slider tail **554** with the distal end of the lockout lever **582** pivots the proximal end of the lockout lever **582** away from the drive member **26** even once the tail **247** of the firing member **240** is no longer acting on the proximal portion of the lockout lever. Accordingly, the drive member **26** and firing member **240** can be distally advanced to fire the staples from the reload. Upon completion of a firing stroke, the slider **550** remains at a distal end of the reload. Thus if the jaw assembly is returned to the open configuration, withdrawing the firing member, the fired reload should be removed and a new unfired reload should be inserted to unlock the reload lockout.

With reference to FIGS. **35-37**, an embodiment of articulation joint **300** to couple the jaw assembly **30** to the distal end of the elongate shaft **20** is illustrated. In the illustrated embodiment, the articulation joint **300** comprises an articulation rod **310** pivotably coupled to the jaw assembly laterally offset from a central longitudinal axis of the shaft assembly. A pivot joint is positioned along the central longitudinal axis. The articulation joint **300** further comprises a support link **320** pivotably coupled to the jaw assembly laterally offset from the central longitudinal axis of the shaft and opposite the articulation rod. The drive beam **26** extends longitudinally along the central longitudinal axis between the articulation rod **310** and the support link **320**. At least a segment of the drive beam **26** extending through the articulation joint **300** is flexible. In some embodiments, the drive beam **26** can be coupled to a flexible segment comprising a stack of shim material, which is flexible while maintaining desired force transmission capabilities for a staple firing operation. The articulation joint can further comprise one or more drive member bearings **330** positioned laterally outwardly of the drive beam **26**. In some embodiments, the drive bearings **330** can comprise a flexible plastic material (FIG. **36**). In other embodiments, the drive bearings **330** can be comprised of a metal shim material (FIG. **37**). Advantageously, the metal shim drive bearing **330** can be keyed into the shaft to provide support to the flexible segment of the drive member. Moreover, the metal shim bearings can have a relatively low profile configuration. The

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metal shim bearings can include a low friction coating such as a TEFLON coating to reduce friction during a firing.

With reference to FIGS. 38A-38B, articulation of the articulation joint to position the jaw assembly in a first articulation position and a second articulation position are illustrated. The articulation rod **310** can be translated proximally (FIG. 38A) or distally (FIG. 38B) relative to the shaft. The lateral offset positioning of the articulation rod **310** articulates the jaw assembly relative to the shaft responsive to translation of the articulation rod. The support link **320** opposite the articulation rod **310** is passive, but can guide articulation motion of the jaw assembly and can advantageously assist in maintaining the flexible portion of the drive beam **26** towards the center of the shaft at the articulation joint, preventing the flexible portion of the drive beam **26** from buckling at the articulated bend at the articulation joint. In other embodiments, the articulation joint can include two articulation rods instead of an articulation rod and support link. In embodiments with two articulation rods, an articulation latch mechanism can be positioned in the shaft to prevent undesired articulation once a staple firing operation has commenced. For example, a latch or brake mechanism can retain the articulation rods from further movement once the drive beam **26** is translated distally.

With reference to FIGS. 39-40, another embodiment of articulation joint **300'** to couple the jaw assembly **30** to the distal end of the elongate shaft **20** is illustrated. The articulation joint **300'** comprises an articulation latch mechanism **340** positioned in the elongate shaft. In the illustrated embodiment, the articulation joint **300'** comprises an articulation rod **310'** pivotably coupled to the jaw assembly laterally offset from a central longitudinal axis of the shaft assembly. A pivot joint is positioned along the central longitudinal axis. The articulation joint **300'** further comprises a support link **320'** pivotably coupled to the jaw assembly laterally offset from the central longitudinal axis of the shaft and opposite the articulation rod. The drive beam **26'** extends longitudinally along the central longitudinal axis between the articulation rod **310'** and the support link **320'**. At least a segment of the drive beam **26'** extending through the articulation joint **300'** is flexible. In some embodiments, the drive beam **26'** can be coupled to a flexible segment comprising a stack of shim material, which is flexible while maintaining desired force transmission capabilities for a staple firing operation. The articulation joint can further comprise one or more drive member bearings **330** positioned laterally outwardly of the drive beam **26'**. In some embodiments, the drive bearings **330** can comprise a flexible plastic material (FIG. 36). In other embodiments, the drive bearings **330'** can be comprised of a metal shim material (FIG. 37). Advantageously, the metal shim drive bearing **330'** can be keyed into the shaft to provide support to the flexible segment of the drive member. Moreover, the metal shim bearings can have a relatively low profile configuration. The metal shim bearings can include a low friction coating such as a TEFLON coating to reduce friction during a firing.

With reference to FIGS. 41-42, articulation of the articulation joint to position the jaw assembly in a first articulation position and a second articulation position are illustrated. The articulation rod **310'** can be translated proximally (FIG. 41) or distally (FIG. 42) relative to the shaft. The lateral offset positioning of the articulation rod **310'** articulates the jaw assembly relative to the shaft responsive to translation of the articulation rod. The support link **320'** opposite the articulation rod **310'** is passive, but can guide articulation motion of the jaw assembly and can advantageously assist in maintaining the flexible portion of the drive beam **26'**

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towards the center of the shaft at the articulation joint, preventing the flexible portion of the drive beam **26'** from buckling at the articulated bend at the articulation joint. In other embodiments, the articulation joint can include two articulation rods instead of an articulation rod and support link.

With reference to FIGS. 39-43, the articulation latch mechanism **340** or brake mechanism of the articulation joint **300'** can retain the articulation rod and support link from further movement once the drive beam **26'** is translated distally. In the illustrated embodiment, the latch mechanism **340** is positioned within the elongate shaft between the proximal end and the distal end thereof. The articulation latching mechanism **340** has an unlatched configuration in which the articulation rod and the support link are slidable within the elongate shaft. Thus, with the articulation latching mechanism in the unlatched configuration, a user can articulate the jaw assembly relative to the elongate shaft by operation of an articulation control on the handle assembly. The articulation latching mechanism **340** further comprises a latched configuration (FIG. 43), wherein the articulation latching mechanism engages the articulation rod and the support link to prevent longitudinal sliding of the articulation link and the support link relative to the elongate shaft. Thus, in the latched configuration, the jaw assembly is retained in an articulated position and the user is prevented from articulating the jaw assembly relative to the elongate shaft.

With continued reference to FIGS. 39-43, in the illustrated embodiment, the articulation latching mechanism **340** comprises a first latch surface, such as a first plurality of teeth **342** formed on the articulation rod **310'**. As illustrated, the first plurality of teeth **342** is positioned within the elongate shaft between the proximal end and the distal end of the articulation rod **310'**. The articulation latching mechanism **340** can further comprise a second latch surface, such as a second plurality of teeth **344** formed on the support link **320'**. As illustrated, in the embodiment of elongate shaft assembly having a latching articulation mechanism, the support link **320'** can extend proximally within the shaft through the articulation latching mechanism **340**. In the illustrated embodiment, the second plurality of teeth **344** is positioned between the proximal end of the support link and the distal end of the support link adjacent the proximal end of the support link **320'**.

In the illustrated embodiment, the articulation latching mechanism **340** further comprises a first shoe **346** having a mating surface such as a first pawl surface **348** formed thereon. The first pawl surface **348** is sized and configured to be engageable with the first plurality of teeth **342**. The first shoe **346** can have a deployment surface opposite the mating surface, the deployment surface is in sliding engagement with the drive beam **26'**. The articulation mechanism **340** can further comprise a second shoe **350** having a mating surface such as a second pawl surface **352** formed thereon. The second pawl surface **352** is sized and configured to be engageable with the second plurality of teeth **344**. The second shoe **350** can have a deployment surface opposite the mating surface, the deployment surface in sliding engagement with the drive beam **26'**. The articulation latching mechanism **340** can further comprise a latching profile formed on the drive beam **26'** between the proximal end and the distal end thereof and positioned within the elongate shaft. In the illustrated embodiment, the drive beam **26'** comprises a recess segment **360** formed therein, a tapered or ramped segment **362** proximal the recess segment, and a latching segment **364** proximal the ramped segment. The

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recess segment **360** has a first width in a direction generally perpendicular to the longitudinal axis of the elongate shaft, and the latching segment **364** has a second width greater than the first width. The articulation latching mechanism can further comprise a biasing member such as a spring clip **370** coupled to the first and second shoes and biasing the shoes **346**, **350** out of engagement with the first and second pluralities of teeth **342**, **344**. The spring clip can also maintain engagement of the deployment surfaces of the shoes **346**, **350** with the latching profile of the drive beam **26'**.

With continued reference to FIGS. **39-43**, in operation, the articulation latching mechanism **340** can initially be positioned in the unlatched configuration (FIGS. **39-42**) such that the jaw assembly can be articulated to a desired orientation relative to the elongate shaft. In this initial positioning, the drive beam **26'** is in a proximal position relative to the elongate shaft, corresponding to an open or partially closed configuration of the jaw assembly. In the unlatched configuration, the first and second shoes **346**, **350** are positioned adjacent the recess segment **360** of the drive beam **26'** in a radially inward position. Once a desired articulated position of the jaw assembly has been selected, a user can proceed to close and fire the jaw assembly, resulting in distal actuation of the drive beam **26'** relative to the elongate shaft. This distal movement of the drive beam **26'** advances the ramped and latching segments **362**, **364** over the deployment surfaces of the first and second shoes **346**, **350**, advancing the shoes radially outwardly. (FIG. **43**). With the first and second shoes **346**, **350** in the radially outward configuration, the first pawl surface **348** engages the first plurality of teeth **342**, and the second pawl surface **348** engages the second plurality of teeth **342** to configure the articulation latch mechanism in the latched configuration. Opening the jaw assembly after a firing sequence will reverse the sequence and return the articulation latch to the unlatched configuration. Thus, desirably, actuation of the drive member **26'** to close and fire the jaw assembly automatically latches an articulated position of the jaw assembly. Advantageously, this latching can reduce or prevent any tendency of the jaw to 'wag' relative to the elongate shaft as the drive beam is advanced around and retracted through the articulation bend. While the illustrated embodiment of actuation latching mechanism includes meshing arrays of teeth on the shoes and actuation rod and support link that define a plurality of discreet latched positions, it is contemplated that in other embodiments, the shoes, actuation rod, and support link can be configured to frictionally engage to define a continuous array of latched articulation positions. Moreover, while the illustrated embodiment includes two shoes each engageable with a corresponding plurality of teeth, in other embodiments, a single shoe can be advanceable to engage a single plurality of teeth on the articulation rod or support link.

With reference to FIGS. **44** and **45A-45D**, a coupler **46** at the distal end of the handle assembly **40** can be coupled to the proximal end of the shaft assembly **20**. The coupler **46** can include a bayonet connection with a lock-in. In the illustrated embodiment, the reload shaft **20** to handle **40** connection comprises a bayonet style connection, in which a user axially aligns and inserts the reload shaft **20** into the handle **40** and rotates the reload shaft **20** approximately 90 degrees to connect. This bayonet connection operatively couples two mechanical functions of the reload shaft **20** to corresponding actuators of the handle **40**. When the bayonet connection is fully coupled, an articulation member within the shaft **20** is coupled to an articulation adapter of the

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handle and a drive member within the shaft **20** is coupled to the actuation adapter. Furthermore, the handle **40** and shaft **20** can be configured with a latch mechanism at the coupler **46** to prevent a user from removing the shaft **20** once the actuation adapter and drive member has been activated. Moreover, the connection at the coupler **46** can include a reload identifying mechanism such that the control system of the handle can detect if a reload shaft is connected, and if so what the attached jaw length of the reload is. It is contemplated that the handle can be used with reload shafts **20** including different length jaw assemblies. In some embodiments the same handle **40** can be used with either 45 mm or 60 mm length jaw assemblies.

In FIG. **45A**, the shaft **20** is positioned in alignment with the coupler **46** on the handle, and a release knob of the coupler **46** is withdrawn to expose a bayonet channel **152** of the coupler **46** on a rotation insert of the coupler **46**. The shaft **20** can include a retention post **22** or boss positionable within the bayonet channel **152**. In the illustrated embodiment, the shaft includes two bosses positioned 180 degrees apart on the outer surface thereof and the coupler **46** includes a corresponding two bayonet channels **152**. It is contemplated that in other embodiments, other numbers and configurations of bosses and bayonet channels can be used to provide a desired connection strength and ease of alignment.

With reference to FIG. **45B**, the retention post **22** of the shaft is positioned within the bayonet channel **152**. With reference to FIG. **45C**, the reload shaft **20** has been rotated 90 degrees relative to the handle such that the retention post **22** of the shaft has reached a connected end of the bayonet channel **152**. With reference to FIG. **45D**, the release knob of the coupler is released to allow a retention recess **154** on the release knob to retain the retention post **22** of the reload shaft **20**.

With reference to FIGS. **46** and **47**, the shaft assembly can include a tubular shaft with the drive member or drive beam **26** and articulation member **206** extending therethrough from the proximal end to the distal end. The drive member can extend generally centrally through the shaft assembly while the articulation member is laterally offset. The proximal end of the tubular shaft can include a coupling collar **410** for coupling to the coupler **46** at the distal end of the handle. In the illustrated embodiment, the shaft assembly can include a proximal shaft 'lock out' mechanism. The lockout mechanism comprises a locking ring positioned within a shaft coupler at the proximal end of the elongate shaft and at least one lockout member radially outwardly advanceable through the coupling collar **410**. The lockout member can be biased radially outwardly, but held in a radially inward position by the locking ring in an initial position. When the proximal end of the shaft is coupled to a handle assembly in a rotation sequence corresponding to a bayonet connection, the locking ring is engaged with a mating surface in the handle assembly and rotates relative to the elongate shaft. This rotation of the locking ring releases the lockout member. Upon removal of the shaft from the handle assembly, the lockout member radially expands. In this expanded position, the lockout member interferes with recoupling the elongate shaft to the handle assembly. Thus, this lockout mechanism can serve to limit inadvertent reuse of an elongate shaft assembly.

With reference to FIGS. **48A** and **48B**, engagement of the bayonet coupling between the shaft assembly and the handle is illustrated. The coupler of the handle can comprise a rotation sleeve for coupling to the coupling collar **410** in which an actuation adapter **124**, an articulation adapter **204**, and an identification sleeve **208** are positioned. During a

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bayonet coupling, the drive member of the shaft engages **26** with the actuation adapter **124**, the articulation member **206** of the shaft engages with the articulation adapter **204**, and a shaft identifier engages with the identification sleeve **208**. FIGS. **49A** and **49B** illustrate the respective engagements with the shaft in a coupled configuration.

With reference to FIGS. **50** and **51**, instead of or in addition to the lockout mechanism described with reference to FIGS. **46** and **47**, certain embodiments of elongate shaft can include a lock-in or retention mechanism that operates upon initial distal advancement of the actuation adapter **124**. As illustrated, a locking member **24** is pivotably coupled to a proximal end of the shaft **20**. The locking member **24** can include a ramped or tapered lock surface at a proximal edge thereof. As illustrated in FIG. **50**, the shaft **20** is in a coupled, but unlocked configuration with respect to the coupler **46**. In the coupled, unlocked configuration, the shaft **20** can be removed from the coupler **46** through the bayonet connection by a reverse of the sequence of operations of FIGS. **45A-45D**. Once the actuation adapter **124** is advancing to operate the stapler, the actuation adapter **124** interacts with the ramped surface of the locking member **24** to advance the locking member radially outward into a locked position. In the locked position (FIG. **51**), the locking member **24** engages a locking ledge on the coupler **46** to lock in the shaft. With the shaft **20** locked in with respect to the handle **40**, the shaft **20** cannot be removed from the handle **40** until the actuation adapter **124** has been returned to a fully proximally retracted position (typically corresponding to a return to a jaws open configuration following a full closure and stapling cycle of the jaw assembly).

Thus, the “lock In” feature prevents a user from removing the shaft from the handle once the drive member **26** has been driven forward. Once the locking member **24** is situated in the slot or ledge of a rotation insert of the coupler **46**, a release knob of the coupler **46** is restricted from being pulled back. This locking action on the coupler prevents the user from rotating the shaft **20** out of the bayonet connection of the coupler **46**.

With reference to FIG. **52**, a proximal end of the shaft assembly comprises a shaft coupler or coupling collar **410** positioned on the proximal end of the tubular shaft. Thus, the stapling system described herein can easily be adapted for use with shaft assemblies having various diameters. In some embodiments, an inner diameter the shaft coupler can be readily resized to accommodate various tubular shafts without requiring different handle assemblies to accommodate shaft assemblies of various diameters.

Although this application discloses certain preferred embodiments and examples, it will be understood by those skilled in the art that the present inventions extend beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. Further, the various features of these inventions can be used alone, or in combination with other features of these inventions other than as expressly described above. Thus, it is intended that the scope of the present inventions herein disclosed should not be limited by the particular disclosed embodiments described above, but should be determined only by a fair reading of the claims.

What is claimed is:

1. A reload assembly for a surgical stapling system, the reload assembly comprising:

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an elongate shaft having a proximal end and a distal end and defining a longitudinal axis extending between the proximal end and the distal end;

a jaw assembly positioned at the distal end of the elongate shaft, the jaw assembly comprising:

a first jaw comprising a reload support configured to receive a staple reload; and

a second jaw pivotably coupled to the first jaw, the second jaw comprising an anvil surface;

an actuation beam longitudinally slidable within the elongate shaft, the actuation beam having a proximal end and a distal end, the distal end of the actuation beam coupled to the jaw assembly; and

a shaft coupler at the proximal end of the elongate shaft, the shaft coupler comprising a locking member positioned therein, a proximal portion of the locking member radially outwardly advanceable relative to the longitudinal axis by distal actuation of the proximal end of the actuation beam to place the shaft coupler in a locked configuration;

wherein the shaft coupler has an unlocked configuration with the locking member in a radially inward position, the shaft removably positionable in a coupler of a handle assembly in the unlocked configuration; and

wherein with the shaft coupler in the locked configuration and the shaft positioned in the coupler of the handle assembly, the shaft is locked in with respect to the handle assembly.

2. The reload assembly of claim 1, wherein with the shaft coupler in the locked configuration the proximal portion of the locking member is radially outwardly advanced.

3. The reload assembly of claim 1, wherein the locking member is pivotably coupled to the elongate shaft and pivotable from an unlocked position to a locked position.

4. The reload assembly of claim 1, wherein the locking member comprises a lever having a proximal end, the locking member comprising a tapered lock surface at the proximal end that is radially outwardly advanceable relative to the longitudinal axis by distal actuation of the actuation beam.

5. The reload assembly of claim 4, wherein proximal retraction of the actuation beam to a fully proximally retracted position returns the shaft coupler to the unlocked configuration.

6. The reload assembly of claim 1, wherein the shaft coupler is configured to couple to the coupler of the handle assembly in a bayonet connection and wherein with the shaft coupler in the locked configuration, a decoupling operation of the bayonet configuration is prevented.

7. The reload assembly of claim 1, wherein the locking member is outwardly advanceable upon initial distal movement of the actuation beam from a fully proximally retracted position.

8. The reload assembly of claim 7, wherein with the actuation beam in the fully proximally retracted position, the jaw assembly is in an open configuration, and wherein the actuation beam is distally longitudinally advanceable a first distance from the fully proximally retracted position to actuate the jaw assembly to a closed configuration.

9. The reload assembly of claim 8, wherein distal longitudinal advancement of the actuation beam distally past the first distance deploys staples from the staple reload received in the reload support.

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